

Kinematics of Rigid Bodies

9 CHAPTER

Key Concepts

- Definition of translation, rotation, and general plane motion
- Governing kinematic equations of rigid body motion
- Governing kinematic equations of fixed axis rotation
- Analysis of motion of two rotating bodies, and motion of one translating and one rotating body
- Study of instantaneous centre of velocity

9.1 INTRODUCTION

A rigid body is idealized as a body that neither deforms nor changes its shape. Formally, it is defined as a collection of particles, having the property that the distance between particles remains unchanged during the motion of the body. Like the approximation of a rigid body as a particle, this is never strictly true. In reality, all bodies deform as they move. However, the approximation remains acceptable as long as the deformations are negligibly small in comparison with the overall motion of the body. As an example, during flight, the flutter of an aircraft wing is obviously negligible relative to the motion of the aircraft as a whole. However, if we are interested in stresses induced in the wing due to flutter, the respective deformations should be taken into account. Here our discussion will remain restricted to the planar motion of rigid bodies. In particular, we will envisage all rigid bodies as thin slabs having motion constrained to lie within the plane of the slab, and we will derive expressions describing the general motion of a rigid body in the plane.

9.2 PROPERTIES OF RIGID BODY MOTION

In the following section, we will identify two properties, velocity and acceleration of the motion of rigid bodies, which simplify the kinematics significantly. In rigid body motion, all points of the body have velocity and acceleration. An arbitrary rigid body motion falls into one of the following three categories:

Translation It is the motion in which every line element within the body moves parallel to its original position. Sometimes it is also called *pure translation*. This motion is of two types, *rectilinear translation* and *curvilinear translation*. In rectilinear motion, each and every point of the body moves along parallel straight lines. On the other hand, all points of a rigid body trace congruent curves during the curvilinear motion of the body. The movements are illustrated in Figs 9.1(a) and (b).

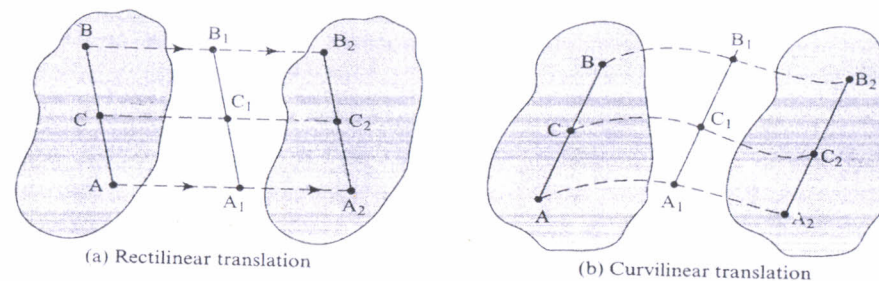


Fig. 9.1

Translational movement is possible along three directions, namely along x -axis, y -axis, and z -axis, provided the motion is described in space. So in space, the maximum possible degree of freedom in translation is three. In planar motion, the maximum possible degree of freedom in translation is two.

Rotation about a fixed axis In *rotation* or *pure rotation*, all particles move in circular paths about the axis of rotation. The motion of the body is completely determined by the angular velocity and acceleration of the rotation. During rotation, each point of the body experiences the same angular velocity and acceleration. As in the case of translation, the degree of freedom in rotation is three in case of spatial motion and two in case of planar motion.

General plane motion If the body lies in the same plane or in parallel plane during motion, it is defined as *plane motion* or *general plane motion*. Any plane motion is neither a pure rotation nor a pure translation. It is basically a combination of pure translation and pure rotation about an axis normal to the plane of motion. It is important to note that in plane motion, *pure translation* and *pure rotation* are commutative. It implies that first rotation and then translation yield the same state of motion, if the rigid body undergoes translation first and then rotation.

We can proceed by demonstrating that every motion of a planar rigid body is associated with a single angular velocity ω and angular acceleration α , describing the angular displacement of an arbitrary line inscribed in the body relative to a fixed direction. Consider a rigid body undergoing plane motion, as shown in Fig. 9.2. The angular positions of two arbitrary lines 1 and 2 attached to the body are specified by θ_1 and θ_2 measured relative to any convenient fixed reference direction. These are related to the intermediate angle β shown as,

$$\theta_2 = \beta + \theta_1 \quad (9.1)$$

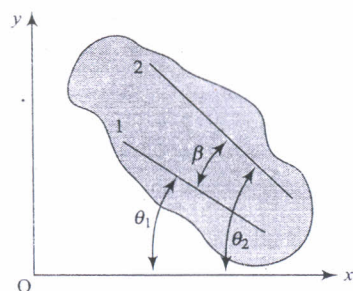


Fig. 9.2

We can observe that as the body is rigid, the distance between each pair of points on the two lines 1 and 2 should be constant, thus angle β must be invariant. Differentiating the above relation, we obtain

$$\frac{d\theta_2}{dt} = \frac{d\theta_1}{dt} = \omega \quad (9.2)$$

$$\text{and } \frac{d^2\theta_2}{dt^2} = \frac{d^2\theta_1}{dt^2} = \alpha \quad (9.3)$$

These hold for all arbitrary lines attached to the body, implying in turn that the body can be associated with a unique angular velocity ω and thereby same angular acceleration α .

Next, consider the motion of a rigid body over the interval t_0 to t_1 as shown in Fig. 9.3, with arbitrary point B_0 taken as reference. Clearly, the motion can be idealized to occur in two stages, namely translation of an arbitrary line A_0B_0 to an intermediate position A_1B_0 and rotation about point B_0 taking A_1B_0 to its final position A_2B_0 . So the effect of the entire planar motion can be decomposed into the effects of a translation and a rotation. While the translational motion is described by the velocity v_B and acceleration a_B of the reference point, the rotational motion is characterized by the unique angular velocity ω and angular acceleration α associated with the body. Thus, we have the property that the motion of a rigid body can be decomposed into a translation of an arbitrary point within the body, followed by a rigid rotation of the body about this point. It is very important to keep in mind that the translation and the rotation should not necessarily follow any order of occurrence. In the present case, first rotation and then translation would yield the same dynamic configuration at t_1 -th instant.

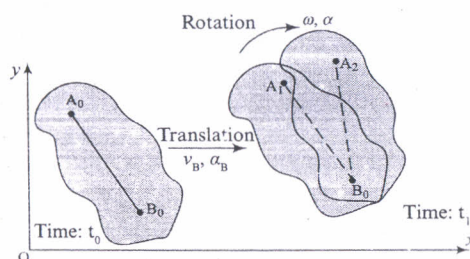


Fig. 9.3

9.3 GOVERNING KINEMATIC EQUATIONS OF RIGID BODY MOTION

With this understanding of the plane motion of rigid bodies, we can derive equations describing the motion. Consider a rigid body having two arbitrarily chosen points A and B moving in a plane with angular velocity ω and angular acceleration α , as shown in Fig. 9.4. We will examine the motion of this body in both the fixed reference O as well as relative to a non-rotating reference attached to point B. We can express the absolute position vector of

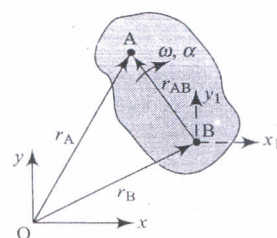


Fig. 9.4

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point A in terms of the absolute position of point B as, $r_A = r_B + r_{AB}$, where r_{AB} is the position vector of A relative to B. An analogous expression for absolute velocities follows by taking time derivatives, $v_A = v_B + v_{AB}$ where $v_{AB} = dr_{AB}/dt$.

Now, as the body moves, point A traces a circular path of radius $r_{AB} = |r_{AB}|$ relative to point B. This is shown in Fig. 9.5. The angular velocity of this motion is simply the angular velocity ω of the rigid body. Thus, we can write

$$v_{AB} = \frac{dr_{AB}}{dt} = \omega \times r_{AB} \quad (9.4)$$

So, the velocity of A is

$$v_A = v_B + \omega \times r_{AB} \quad (9.5)$$

which is basically the vector sum of a translational component v_B and a rotational component $\omega \times r_{AB}$.

In order to derive the expressions for the acceleration of an arbitrary point of a rigid body, differentiate the equation for velocities to obtain $a_A = a_B + a_{AB}$. Here, a_{AB} is the acceleration of point A relative to the reference at B and

$$a_{AB} = \frac{d}{dt} (\omega \times r_{AB}) = \frac{d\omega}{dt} \times r_{AB} + \omega \times \frac{dr_{AB}}{dt} = \alpha \times r_{AB} + \omega \times (\omega \times r_{AB}) \quad (9.6)$$

$$\text{So, } a_A = a_B + \alpha \times r_{AB} + \omega \times (\omega \times r_{AB}) \quad (9.7)$$

Thus, like the absolute velocity, the absolute acceleration of point A is the vector sum of a translational part a_B and a rotational part

$$a_{AB} = \alpha \times r_{AB} + \omega \times (\omega \times r_{AB})$$

Alternatively, we can write the equations of motion as,

$$v_A = v_B + \omega r_{AB}$$

$$a_A = a_B + \alpha r_{AB} + \omega^2 r_{AB} = a_B + (a_{AB})_t + (a_{AB})_n \quad (9.8)$$

Here the rotational component of acceleration is further reduced to tangential and normal components of a_{AB} . Decomposition of absolute velocity and absolute acceleration of point A of the rigid body is shown in Figs 9.6 and 9.7, respectively, along with vector diagrams.

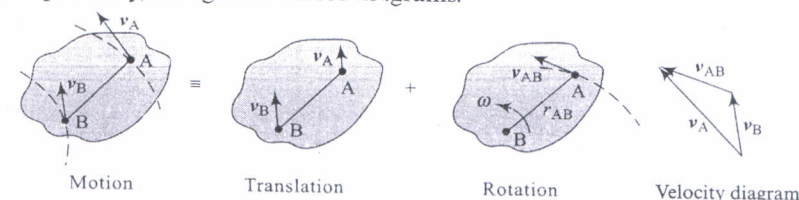


Fig. 9.6

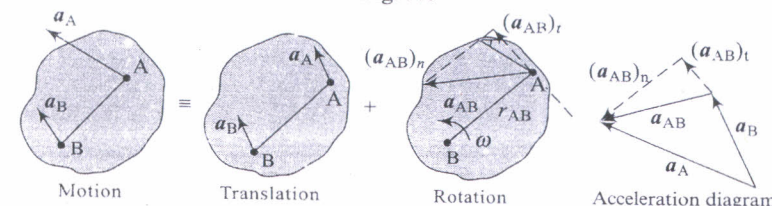


Fig. 9.7

9.4 FIXED AXIS ROTATION OF A RIGID BODY

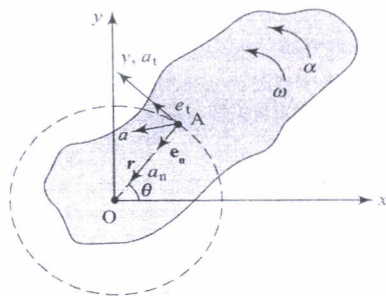


Fig. 9.8

In this section, we will apply the derived kinematic relations to the case of a rigid body rotation about a fixed axis. As will be seen, the relations will get reduced to familiar forms, once n - t coordinates are introduced. Consider an arbitrary point A of a rigid body rotating with angular velocity ω and angular acceleration α about the axis O. Let e_n and e_t be the unit vectors at point A as shown in Fig. 9.8, where e_t is the tangent and e_n is the normal to the path of point A.

Then, using the kinematic relations for

general rigid body motion with axis O taken as reference, we obtain the expressions for velocity v and acceleration a of point A as

$$\begin{aligned} v &= \omega \times r \\ a &= \alpha \times r + \omega \times (\omega \times r) \end{aligned} \quad (9.9)$$

Here r is the position vector of A relative to the axis. Now, as the motion is planar, we have,

$$v = v_t e_t = \omega r e_t \quad (9.10)$$

$$a = a_n e_n + a_t e_t = (\omega^2 r) e_n + (\alpha r) e_t \quad (9.11)$$

Here the velocity of a point in the body is perpendicular to the line connecting the point to the axis of rotation, with magnitude proportional to its distance from the axis. The acceleration, on the other hand, is composed of two parts: a_n can be recognized as the normal acceleration or centripetal acceleration directed towards the axis and a_t is the tangential acceleration.

9.5 CONTACT OF TWO ROTATING RIGID BODIES

An example of this form of contact is that between gears in a gear train like spur, bevel, helical, or worm gears. However, in our consideration of planar motion, we are limiting the analysis to contact between gears that have a common axis of rotation.

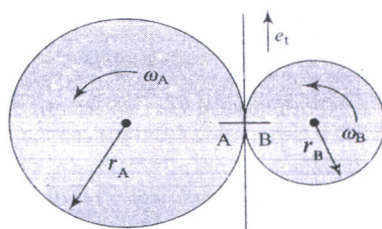


Fig. 9.9

$$v_A = v_B$$

Consider the motion of instantaneous contact points A and B, each on one of a pair of interlocking gears, as shown in Fig. 9.9. Assume that no sliding occurs at the contact surfaces. As contact takes place without slipping, the velocity of A relative to B must vanish. If r_A and r_B are the radii, v_A and v_B are the absolute velocities, and ω_A and ω_B are the angular velocities of gears A and B, respectively, then we can write

$$(9.12)$$

$$\begin{aligned} \text{or } r_A \omega_A e_{At} &= r_B \omega_B e_{Bt} \\ \text{However, as } e_{At} &= -e_{Bt} = e_t \text{ we have,} \\ r_A \omega_A &= -r_B \omega_B \end{aligned}$$

$$\text{or } \frac{\omega_A}{\omega_B} = -\frac{r_B}{r_A} \quad (9.13)$$

The relationship suggests that the ratio of angular velocities of a pair of circular rotating bodies in contact is inversely proportional to the ratio of their radii. Taking time derivatives, we obtain a relationship between the accelerations of points A and B as,

$$r_A \alpha_A = -r_B \alpha_B$$

$$\text{or } \frac{\alpha_A}{\alpha_B} = -\frac{r_B}{r_A} \quad (9.14)$$

This implies that the two contact points A and B have tangential accelerations of equal magnitude.

9.6 CONTACT OF A ROTATING RIGID BODY WITH A TRANSLATING RIGID BODY

Contact of this form is encountered in belts, ropes, and chain drives of all kinds, as well as between rack and pinion.

As before, consider the motion of instantaneous contact points, A on the rotating body and B on the translating body as shown in Fig. 9.10. As the two points must have identical absolute velocities, the velocity of point B must be directed along e_t , tangent to the path of A. So the mutual relationships of velocity and accelerations are

$$r_A \omega_A = v_B \quad (9.15)$$

$$r_A \alpha_A = a_B \quad (9.16)$$

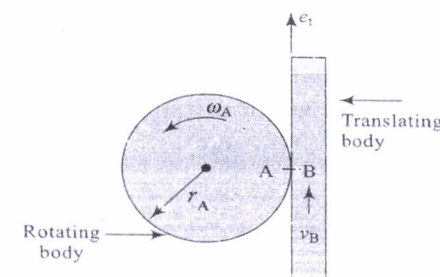


Fig. 9.10

9.7 INSTANTANEOUS CENTRE OF ROTATION

Considering a general plane motion of a body, at a given instant, the velocities of various particles of the body could be expressed as the result of a pure rotation whose axis is perpendicular to the plane. This axis intersects the plane at a point called the *instantaneous centre of rotation*. Sometimes this point more precisely is referred to as the *instantaneous centre of velocity*.

The position of this particular point can be defined in many ways. Suppose the directions of the velocities of two particles A and B of a rigid body are known,

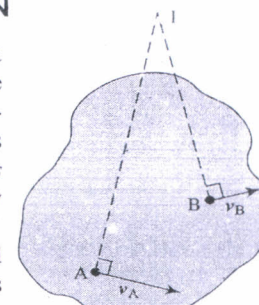


Fig. 9.11

their lines of action are non-parallel and their magnitudes are different, as shown in Fig. 9.11. Draw the perpendicular to the line of action of v_A through A and the perpendicular to the line of action of v_B through B. Locate the point of intersection of these two perpendiculars. This is the instantaneous centre of rotation (I) of the body.

Another case is that the line of action of v_A through A and the line of action of v_B through B are parallel but their known magnitudes are different, as shown in Fig. 9.12. In this case, draw two lines joining the two extremities of v_A and v_B . The point of intersection of these two lines will give the instantaneous centre of rotation (I). We can further illustrate this concept of centre of instantaneous velocity with an example as below.

Consider a bar AB leaning against the wall and slipping downward, as shown in Fig. 9.13. It is clear that while the bar is in contact with the wall and the floor, the velocity at point A will be along the vertical direction, whereas the velocity at point B will be along the horizontal direction. Therefore, drawing the perpendiculars to the lines of actions of v_A and v_B through points A and B, and locating their point of intersection, we can determine the instantaneous centre of rotation (I). It should be noted that, for a general motion, the location of the centre of rotation would change in time. The path described by the instantaneous centre of rotation is called the *space centrode*, and the locus of the positions of the instantaneous centres on the body is called the *body centrode*. At a given instant, the space centrode and the body centrode curves are tangent. The *tangency point* is precisely the instantaneous centre of rotation (I). It is not difficult to show that, for the present example, the space and body centrodes

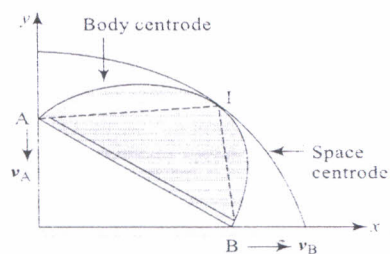


Fig. 9.13

are circular arcs, assuming that the points A and B remain in contact with the walls at all times. From this example, it should be clear that although we think about the instantaneous centre of rotation as a point attached to the body, it need not be a material point. In fact, it can be a point *outside* the body. It is also possible to consider the *instantaneous centre of acceleration* as the point at which the instantaneous acceleration is zero.

RECAPITULATION

- Translation is of two types, rectilinear translation and curvilinear translation. General plane motion is a combination of translation and rotation, irrespective of the order of occurrence. General plane motion of a rigid body is associated with a unique angular velocity ω and same angular acceleration α .

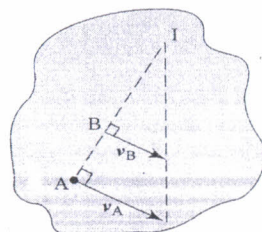


Fig. 9.12

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- Governing equations of velocity and acceleration of rigid body motion are $v_A = v_B + \omega r_{AB}$ and $a_A = a_B + \alpha r_{AB} + \omega^2 r_{AB} = a_B + (a_{AB})_t + (a_{AB})_n$.
- In fixed axis rotation, the velocity and acceleration of a point A in the rigid body can be expressed as $v = v_t e_t = \omega r e_t$ and $a = a_n e_n + a_t e_t = (\omega^2 r) e_n + (\alpha r) e_t$.
- For two rotating bodies in contact, the kinematic relationships between the contact points A and B are $\frac{\omega_A}{\omega_B} = -\frac{r_B}{r_A}$ and $\frac{\alpha_A}{\alpha_B} = -\frac{r_B}{r_A}$ where the symbols carry usual meaning.
- In the motion of a rotating body and a translating body, the kinematic relationships between the two contact points are $r_A \omega_A = v_B$ and $r_A \alpha_A = a_B$.
- In the planar motion of a rigid body, if a point I is found such that at any instant at that particular point the velocity is zero, the point is defined as instantaneous centre of rotation or instantaneous centre of velocity. Velocity of the rigid body at that particular instant would consist only of a rotation of the body about instantaneous of rotation. This point may or may not lie within the body.

NUMERICAL EXAMPLES

Example 9.1 At a given instant, a shaft is rotating at 50 rpm about a fixed axis and 20 s later, it is rotating at 1050 rpm. Determine the average angular acceleration in rad/s^2 .

Solution Here initial velocity $\omega_0 = \frac{2\pi \times 50}{60} \text{ rad/s}$

Final velocity $\omega_1 = \frac{2\pi \times 1050}{60} \text{ rad/s}$

Using equation of motion, $\omega_1 = \omega_0 + \alpha t$

$$\text{or } \alpha = \frac{\omega_1 - \omega_0}{t} = \frac{2\pi}{60} (1050 - 50) \times \frac{1}{20} = 5.236 \text{ rad/s}^2$$

Example 9.2 A flywheel of diameter 50 cm starts from rest with constant angular acceleration of 2 rad/s^2 . Determine the tangential and the normal components of acceleration of a point on its rim 3 s after the motion began.

Solution Here, initial velocity $\omega_0 = 0$ and constant acceleration $\alpha = 2 \text{ rad/s}^2$.

After 3 s of start of motion, the angular velocity will be

$$\omega = \omega_0 + \alpha t = 2 \times 3 = 6 \text{ rad/s}$$

Hence, normal acceleration, $a_n = \omega^2 r = 6^2 \times \frac{0.50}{2} = 9 \text{ m/s}^2$

and tangential acceleration, $a_t = r\alpha = \frac{0.50}{2} \times 2 = 0.5 \text{ m/s}^2$

Example 9.3 A particle is at rest on a phonograph turntable at a radius R from the centre. If the turntable starts from rest and rotates with uniform acceleration C , what are the expressions of tangential and normal acceleration of the particle at time T ?

Solution At time T , the angular velocity of turntable, $\omega = 0 + CT = CT$

So, tangential acceleration, $a_t = CR$

and normal acceleration, $a_n = \omega^2 R = C^2 T^2 R$

Example 9.4 A rotor decreases uniformly from rotating speed of 1800 rpm to rest in 320 s. Determine its angular deceleration and the number of radians rotated before coming to rest.

Solution Here, $\omega_0 = \frac{2\pi \times 1800}{60}$ rad/s, $\omega_1 = 0$, and $t = 320$ s

Using the relation, $\omega_1 = \omega_0 - \alpha t$,

$$0 = \frac{2\pi \times 1800}{60} - 320 \alpha$$

So, $\alpha = 0.589$ rad/s²

Now, the amount of rotation before coming to rest,

$$\begin{aligned} \theta &= \omega_0 t - \frac{1}{2} \alpha t^2 \\ &= \frac{2\pi \times 1800}{60} \times 320 - \frac{1}{2} \times 0.589 \times 320^2 \\ &= 30161.78 \text{ rad} \end{aligned}$$

Example 9.5 A bar pivoted at one end moving 5 rad/s clockwise is subjected to a constant angular deceleration. After a certain time interval, the bar has an angular displacement of 8 rad counterclockwise and has moved through a total angle of 20.5 rad. What is the angular velocity at the end of the time interval?

Solution Here, the total angular movement of 20.5 rad has one part 8 rad counterclockwise and remaining $(20.5 - 8) = 12.5$ rad clockwise. Starting from movement with speed 5 rad/s, the bar will come momentarily at rest and then it will start to move counterclockwise, if the constant angular acceleration is α , for the first part of motion, i.e., along clockwise. Using

$$0 = \omega^2 - 2\alpha\theta$$

$$\text{or } \alpha = \frac{\omega^2}{2\theta} = \frac{5^2}{2 \times 12.5} = 1 \text{ rad/s}^2$$

For the first part of motion, this α acted as deceleration. But for the second part of motion, α will be acceleration, initial speed will be zero, and angle traversed 8 rad. So the rotational speed after traversing 8 rad angle will be

$$\omega_1^2 = 0 + 2\alpha\theta = 0 + 2 \times 1 \times 8$$

$$\text{or } \omega_1 = 4 \text{ rad/s}$$

Example 9.6 The drum shown in Fig. E9.6 having diameter 800 mm is used to hoist the weight from a depth of 8 m. The drum accelerates uniformly from rest to 15 rpm in 1.5 s, and then moves at a constant speed of 15 rpm. What is the total elapsed time?

Solution Final rotational velocity $\omega = 2\pi \times 15/60$ rad/s. Starting from rest to achieve 15 rpm speed, the drum rotates by an amount θ and acceleration α . So,

$$\frac{2\pi \times 15}{60} = 0 + \alpha \times 1.5$$

$$\text{or } \alpha = \frac{\pi}{3} \text{ rad/s}^2$$

$$\text{and } \theta = 0 + \frac{1}{2} \times \frac{\pi}{3} \times 1.5^2 = 0.375 \pi$$

For a full rotation of drum, i.e., for 2π rad rotation,
Linear distance moved $= (2\pi \times 0.8) \text{ m} = 5.027 \text{ m}$

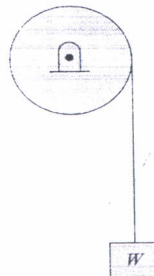


Fig. E9.6

Rest of the height, the weight is to move $= (8 - 5.027) \text{ m} = 2.973 \text{ m}$. This distance is equivalent to the rotation of $(2.973/0.8) \text{ rad} = 3.71625 \text{ rad}$. This angular movement will be done at a constant speed of 15 rpm. So, the time required to move 3.71625 rad angle is $3.71625 \times 60/(2\pi \times 15) = 2.37 \text{ s}$. Now, the total time required $= 1.5 + 2.37 = 3.87 \text{ s}$.

Example 9.7 Two balls are moving with uniform velocities along x - and y -axes towards the origin, as shown in Fig. E9.7. After time t of their journey, they reach positions P_1 and Q_1 and these distances of traverse are x and y , respectively. The initial positions of the balls from the origin O are designated by distances x_P and y_Q , respectively. Determine the angular velocity of the line joining P_1 and Q_1 and show that angular

velocity will be maximum for $t = \frac{ux_P + vy_Q}{u^2 + v^2}$.

Solution After time t , the distances traversed by the balls are

$$x = ut = PP_1 \text{ and } y = vt = QQ_1$$

$$\text{Now, } OP_1 = OP - PP_1 = x_P - ut$$

$$OQ_1 = OQ - QQ_1 = y_Q - vt$$

If the inclination of line P_1Q_1 is θ ,

$$\tan \theta = \frac{OQ_1}{OP_1} = \frac{y_Q - vt}{x_P - ut} \quad (1)$$

Differentiating Eq. (1) with respect to t , we obtain

$$\sec^2 \theta \frac{d\theta}{dt} = \frac{y_Q - vx_P}{(x_P - ut)^2}$$

$$\text{or } \frac{d\theta}{dt} = \frac{1}{\sec^2 \theta} \frac{y_Q - vx_P}{(x_P - ut)^2}$$

$$= \frac{1}{(1 + \tan^2 \theta)} \frac{y_Q - vx_P}{(x_P - ut)^2} = \frac{1}{1 + \left(\frac{y_Q - vt}{x_P - ut}\right)^2} \frac{y_Q - vx_P}{(x_P - ut)^2}$$

$$\text{Hence } \omega = \frac{y_Q - vx_P}{(x_P - ut)^2 + (y_Q - vt)^2} \quad (2)$$

For ω as in Eq. (2) to be maximum, $[(x_P - ut)^2 + (y_Q - vt)^2]$ should be minimum.

$$\text{So, } \frac{d}{dt} [(x_P - ut)^2 + (y_Q - vt)^2] = 0$$

$$\text{or } 2(x_P - ut)(-u) + 2(y_Q - vt)(-v) = 0$$

$$\text{or } -ux_P - vy_Q + (u^2 + v^2)t = 0$$

$$\text{So, } t = \frac{ux_P + vy_Q}{u^2 + v^2}$$

Example 9.8 The armature of an electric motor has angular speed $N = 1800$ rpm at the instant when the power is cut off. If it comes to rest in 6 s, calculate the angular deceleration α assuming that it is constant. How many complete revolutions does the armature make during this period?

Solution Initial rotational speed $\omega_0 = \frac{2\pi \times 1800}{60} = 60\pi$ rad/s

Applying the relation, $\omega = \omega_0 - \alpha t$, we get
 $0 = 60\pi - \alpha \times 6$

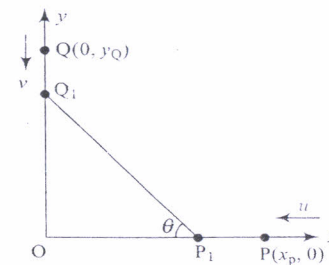


Fig. E9.7

So, $\alpha = 10\pi \text{ rad/s}^2$

Total amount of rotation,

$$\begin{aligned}\theta &= \omega_0 t - \frac{1}{2} \alpha t^2 \\ &= 60\pi \times 6 - \frac{1}{2} \times 10\pi \times 6^2 = 180\pi \text{ rad}\end{aligned}$$

As a complete revolution indicates rotation of 2π rad, number of complete revolutions before coming to rest will be $180\pi/2\pi = 90$.

Example 9.9 A flywheel of 550 mm diameter is brought uniformly from rest up to a speed of 350 rpm in 20 s. Find the velocity and the acceleration of a point on its rim 3 s after starting from rest.

Solution Initial speed $\omega_0 = 0$

Speed at 20 s time, $\omega = \frac{2\pi \times 350}{60} = \frac{35\pi}{3} \text{ rad/s}$

So, acceleration $\alpha = \frac{\omega - \omega_0}{t} = \frac{35\pi}{3 \times 20} \text{ rad/s}^2$

Now, speed of the mentioned point, $\omega_t = \omega_0 + \alpha t = \frac{35\pi}{3 \times 20} \times 3 = \frac{7\pi}{4} \text{ rad/s}$

So, velocity of the point, $v = r\omega_t = \frac{0.55}{2} \times \frac{7\pi}{4} = 1.512 \text{ m/s}$

Tangential acceleration, $a_t = r\alpha = \frac{0.55}{2} \times \frac{35\pi}{3 \times 20} = 0.504 \text{ m/s}^2$

and normal acceleration, $a_n = \omega_t^2 r = \left(\frac{7\pi}{4}\right)^2 \times \frac{0.55}{2} = 8.312 \text{ m/s}^2$

So, total acceleration, $a = \sqrt{a_t^2 + a_n^2} = 8.3273 \text{ m/s}^2$

Example 9.10 In the torsional pendulum shown in Fig. E9.10, the angle of rotation measured from equilibrium position is given as $\theta = \theta_0 \cos pt$. Derive the general expressions for the angular velocity and components of angular acceleration of the disk. Determine the maximum values of these components if the disk makes 2 oscillations per second having the radius $r = 12 \text{ cm}$ and maximum angular velocity $2\pi \text{ rad/s}$.



Fig. E9.10

Solution Here $\theta = \theta_0 \cos pt$

Differentiating once, $\omega = \dot{\theta} = -p\theta_0 \sin pt$ (1)

Further differentiating, $\alpha = \dot{\omega} = -p^2\theta_0 \cos pt$ (2)

Now, tangential acceleration, $a_t = r\alpha = -p^2 r \theta_0 \cos pt$ (3)

Normal acceleration, $a_n = \omega^2 r = p^2 r \theta_0^2 \sin^2 pt$ (4)

Given the time period of oscillation $\tau = \frac{2\pi}{p} = \frac{1}{2}$

So, $p = 4\pi$

Using Eq. (1) for maximum angular velocity,

$$2\pi = -(4\pi\theta_0) \sin(4\pi t) \quad (5)$$

Now, the value of $\sin(4\pi t)$ will be such that, $2\pi = \pm(4\pi\theta_0)$

or $\theta_0 = \pm \frac{1}{2}$

Substituting the values in Eqs (3) and (4), we obtain the components of maximum acceleration as

$$a_{t_{\max}} = (4\pi)^2 \times \frac{1}{2} \times 0.12 = 9.475 \text{ m/s}^2$$

$$a_{n_{\max}} = (4\pi)^2 \times 0.12 \times \left(\frac{1}{2}\right)^2 = 4.737 \text{ m/s}^2$$

Example 9.11 A slender, rigid, semicircular wire of radius R is supported in its own vertical plane by a hinge O and a peg P , as shown in Fig. E9.11. If the peg starts moving from O with constant speed u along the direction x , determine the angular velocity of the wire at the instant when $\theta = 60^\circ$.

Solution From geometry of the figure, $\sin \theta = \frac{ut}{2R}$ (1)

Differentiating the above equation with respect to time,

$$\cos \theta \frac{d\theta}{dt} = \frac{u}{2R}$$

or $\frac{d\theta}{dt} = \frac{1}{\cos \theta} \frac{u}{2R} = \frac{1}{\left[1 - \left(\frac{ut}{2R}\right)^2\right]^{1/2}} \frac{u}{2R}$

or $\omega = \frac{u}{\sqrt{4R^2 - u^2 t^2}} \quad (2)$

At $\theta = 60^\circ$, from Eq. (1)

$$\sin 60^\circ = \frac{ut}{2R}$$

or $t = \frac{R\sqrt{3}}{u}$

Substituting this value of t in Eq. (2), we get

$$\omega = \frac{u}{\left(4R^2 - u^2 \times \frac{3R^2}{u^2}\right)^{1/2}} = \frac{u}{R}$$

Example 9.12 A slotted bar OQ hinged at O and supported by a pin $P(x, y)$ which slides freely along the slot as the block to which pin is attached slides along the horizontal plane. If at zeroth time instant, the bar is vertical and the block moves rightward, derive the general expressions for angular velocity and angular accelerations of bar PQ .

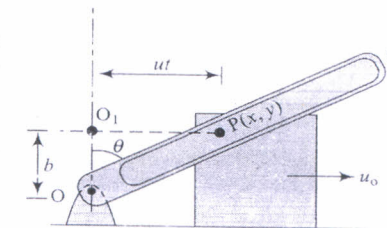


Fig. E9.12

Solution At any time instant, the location of the bar is shown in Fig. E9.12, where θ is the included angle. Using geometry, we have,

$$\tan \theta = \frac{O_1 P}{OO_1} = \frac{ut}{b} \quad (1)$$

Differentiating once with respect to time, we get

$$\sec^2 \theta \frac{d\theta}{dt} = \frac{u}{b}$$

$$\text{or } \frac{d\theta}{dt} = \frac{1}{\sec^2 \theta} \frac{u}{b}$$

$$= \frac{1}{(1 + \tan^2 \theta)} \frac{u}{b} = \frac{1}{\left(1 + \frac{u^2 t^2}{b^2}\right)} \frac{u}{b}$$

$$\omega = \frac{ub}{(b^2 + u^2 t^2)} \quad (2)$$

Differentiating Eq. (2) with respect to time, we obtain,

$$\alpha = \frac{d\omega}{dt} = \frac{(b^2 + u^2 t^2) 0 - ub(2u^2 t)}{(b^2 + u^2 t^2)^2} = -\frac{2u^3 bt}{(b^2 + u^2 t^2)^2} \quad (3)$$

Example 9.13 Considering the system in the previous example, determine the value of θ for which negative angular acceleration $\ddot{\theta}$ of the bar OQ has its maximum value.

Solution From the previous example, $\theta = \tan^{-1} \left(\frac{ut}{b} \right)$ (1)

$$\ddot{\theta} = -\frac{2u^3 bt}{(b^2 + u^2 t^2)^2} \quad (2)$$

For negative value of $\ddot{\theta}$ to be maximum,

$$\frac{d}{dt} \left[\frac{t}{(b^2 + u^2 t^2)^2} \right] = 0$$

$$\text{or } \frac{(b^2 + u^2 t^2)^2 - t \times 2(b^2 + u^2 t^2) \times 2u^2 t}{(b^2 + u^2 t^2)^4} = 0$$

$$\text{or } b^2 + u^2 t^2 - 4u^2 t^2 = 0$$

$$\text{So } t = \frac{1}{\sqrt{3}} \frac{b}{u}$$

Substituting the value of t in Eq. (1), we get

$$\theta = \tan^{-1} \left(\frac{u}{b} \frac{1}{\sqrt{3}} \frac{b}{u} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = 30^\circ$$

So, the required value of θ is 30° .

Example 9.14 A flywheel rotating freely at the speed of 1750 rpm clockwise is provided with a counterclockwise torque, which is first applied at time $t = 0$ producing a counterclockwise acceleration $\alpha = (4.5t) \text{ rad/s}^2$. Determine the (i) time required to produce clockwise angular speed 950 rpm, (ii) time required to reverse the direction of rotation, (iii) total number of revolutions during the first 14 s of movement.

Solution As the rotational speed is clockwise and acceleration counterclockwise, it will virtually behave as deceleration till the flywheel comes to rest. After that instant, the flywheel will move counterclockwise with rotational acceleration counterclockwise.

$$(i) \text{ Here, } \omega_0 = \frac{2\pi \times 1750}{60} \text{ rad/s} \quad \text{and} \quad \alpha_t = \frac{2\pi \times 950}{60} \text{ rad/s}$$

Using the relation, and substituting the values, we get $\omega_t = \omega_0 - \alpha t$

Putting the values, we get

$$\frac{2\pi \times 950}{60} = \frac{2\pi \times 1750}{60} - 4.5t \times t$$

$$\text{or } t = 4.315 \text{ s}$$

(ii) During the instant of reversal of direction of rotation, the rotational speed of the flywheel will be momentarily zero. So,

$$0 = \frac{2\pi \times 1750}{60} - 4.5t \times t$$

$$\text{or } t = 6.382 \text{ s}$$

(iii) During the first 6.382 s of rotation in the clockwise direction,

$$\text{Total angle rotated, } \theta_1 = \omega_0 \times 6.382 - \frac{1}{2} \times (4.5 \times 6.382) \times (6.382)^2$$

$$= 584.7 \text{ rad (clockwise)}$$

In the remaining $(14 - 6.382) \text{ s}$, i.e., 7.618 s, the flywheel will make counterclockwise rotation, starting from rest with acceleration $(4.5t) \text{ rad/s}^2$. So, the amount

$$\text{of rotation } \theta_2 = 0 + \frac{1}{2} \times (4.5 \times 7.618) \times (7.618)^2$$

$$= 994.73 \text{ rad (counterclockwise)}$$

Hence the total rotation, $\theta = \theta_1 + \theta_2$

$$= (584.7 + 994.73) \text{ rad} = 1579.43 \text{ rad}$$

$$= \frac{1579.43}{2\pi} \text{ rev} = 251.374 \text{ revolutions}$$

Example 9.15 A string is wound around a pulley of 0.28 m diameter. One of its ends is fixed to the pulley while the other end is fixed to a weight W , as shown in Fig. E9.15. This weight moves a distance of 7.7 m starting from rest in 4.3 s. Determine the angular speed of the pulley. Also determine the total length traversed by the weight to make the pulley rotate by 425 rpm.

Solution For weight: initial velocity $u = 0$, distance travelled $S = 7.7 \text{ m}$, time required $t = 4.3 \text{ s}$. Using the relationship,

$$S = ut + \frac{1}{2} at^2$$

$$\text{or } 7.7 = \frac{1}{2} a \times (4.3)^2 \text{ or } a = 0.833 \text{ m/s}^2$$

So, the angular acceleration of pulley,

$$\alpha = \frac{0.833}{R} = \frac{0.833}{\left(\frac{0.28}{2}\right)} = 5.95 \text{ rad/s}^2$$

Considering that the rotation of pulley starts from rest, the rotational speed of pulley after this time interval will be,

$$\omega = 0 + \alpha \times 4.3 = 5.95 \times 4.3 = 25.585 \text{ rad/s}$$

$$\text{Final speed of pulley, } \omega_f = \frac{2\pi \times 425}{60} \text{ rad/s}$$

If the rotation continues with constant acceleration, time required to reach the final

$$\text{speed} = \frac{\omega_f - 0}{\alpha} = \frac{2\pi \times 425}{60 \times 5.95} = 7.48 \text{ s}$$

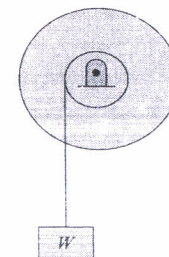


Fig. E9.15

Now, consider linear movement of weight where initial velocity is zero and acceleration 0.833 m/s^2 constant. So, the total distance traversed by the weight

$$= \frac{1}{2} \times 0.833 \times (7.48)^2 = 23.303 \text{ m}$$

Example 9.16 A pinion P and gear G are attached for hoisting a load W , as shown in Fig. E9.16. The radii of pinion and gear are 95 mm and 350 mm, respectively. The load is lifted from anywhere and after passing a distance 0.73 m acquires a velocity 2.1 m/s, with constant acceleration. Determine (i) the acceleration of the point A on the cable in contact with the gear and (ii) the angular speed and angular acceleration of pinion P.

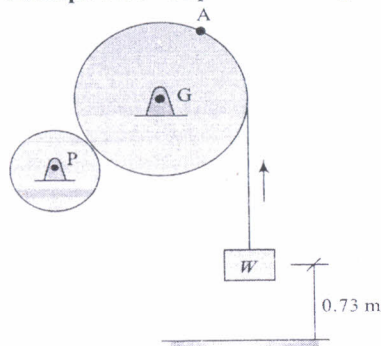


Fig. E9.16

Solution To determine the upward acceleration of the block,

$$v^2 = 0^2 + 2aS \quad \text{or} \quad a = \frac{v^2}{2S} = \frac{(2.1)^2}{2 \times 0.73} = 3.02 \text{ m/s}^2$$

So, normal acceleration $a_n = \frac{v^2}{r_G} = \frac{(2.1)^2}{0.35} = 12.6 \text{ m/s}^2$

Tangential acceleration $a_t = a = 3.02 \text{ m/s}^2$

At point A, the total acceleration

$$a_A = \sqrt{a_n^2 + a_t^2} = \sqrt{12.6^2 + 3.02^2} = 12.957 \text{ m/s}^2$$

Now, angular speed of gear G, $\omega_G = \frac{v}{r_G} = \frac{2.1}{0.35} = 6 \text{ rad/s}$

Angular acceleration of gear G, $\alpha_G = \frac{a_t}{r_G} = \frac{3.02}{0.35} = 8.63 \text{ rad/s}^2$

From the condition of compatibility, at the point of contact of pinion and gear,

$$\omega_P r_P = \omega_G r_G \quad (1)$$

$$\text{and} \quad \alpha_P r_P = \alpha_G r_G \quad (2)$$

Using Eq. (1),

$$\omega_P = \omega_G \times \frac{r_G}{r_P} = 6 \times \frac{0.350}{0.095} = 22.105 \text{ rad/s}$$

Similarly, from Eq. (2),

$$\alpha_P = \alpha_G \times \frac{r_G}{r_P} = 8.63 \times \frac{0.350}{0.095} = 31.795 \text{ rad/s}^2$$

Example 9.17 A wheel of radius R rolls on a flat surface without slipping, at constant angular acceleration (Fig. E9.17). Study the angular motion of the wheel in terms of the linear motion of the centre C. Also determine the acceleration of a point P on the rim of the wheel as it comes into contact with the surface on which the wheel rolls.

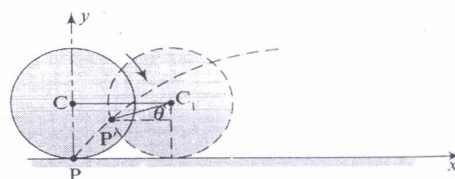


Fig. E9.17

Solution At a certain time instant, the location of the wheel is shown in dotted line in Fig. E9.17. Here linear shifting of centre is CC_1 . The point P got shifted to P' being rotated by an amount θ .

Here $CC_1 = S = R\theta$, where θ is measured in radians. Now, linear velocity of the movement of centre C is $v = \frac{dS}{dt} = R \frac{d\theta}{dt} = R\omega$, where ω is the rotational speed of the wheel.

Again linear acceleration, $a = \frac{dv}{dt} = R \frac{d\omega}{dt} = R\alpha$, where α is the angular acceleration.

Let us assume the coordinates of P' are (x, y) . Using simple geometry,

$$x = CC_1 - R \sin \theta = S - R \sin \theta = R\theta - R \sin \theta \quad (1)$$

$$y = R - R \cos \theta \quad (2)$$

Linear velocity and acceleration along the x -axis will be obtained by differentiating Eq. (1) once and twice, respectively, with respect to time. So,

$$\dot{x} = R\dot{\theta} - R\dot{\theta} \cos \theta = R\omega(1 - \cos \theta) = v(1 - \cos \theta)$$

$$x = R\ddot{\theta} - R\ddot{\theta} \cos \theta + R\dot{\theta} \sin \theta \cdot \dot{\theta} = R\ddot{\theta}(1 - \cos \theta) + (R\dot{\theta})^2 \times \frac{1}{R} \sin \theta$$

$$= a(1 - \cos \theta) + \frac{v^2}{R} \sin \theta$$

Similarly, for motion along the y -axis, velocity and acceleration will be

$$\dot{y} = R \sin \theta \dot{\theta} = R\dot{\theta} \sin \theta = v \sin \theta$$

$$\ddot{y} = R \cos \theta (\dot{\theta})^2 + R \sin \theta \ddot{\theta} = R\ddot{\theta} \sin \theta + \frac{(R\dot{\theta})^2}{R} \cos \theta$$

$$= a \sin \theta + \frac{v^2}{R} \cos \theta$$

Example 9.18 In the gear system, shown in Fig. E9.18, gear wheels G_1 and G_2 are connected. A pulley of radius r is attached with gear G_2 . A load W is made to move with a rope wound around the pulley. If the load starts from rest and moves upward with constant acceleration, derive the equation of motion of gear G_1 .

Solution Let us assume that the initial velocity of W is zero and constant acceleration is a . After time interval t , velocity of load

$$= 0 + at = at$$

So, peripheral velocity of pulley also will be at . Considering angular speed of the pulley ω_2 , we can write,

$$\omega_2 r = at$$

$$\text{or} \quad \omega_2 = \frac{at}{r} \quad (1)$$

Again for gear wheel G_2 ,

$$\omega_2 R_2 = \frac{at}{r} R_2 \quad (2)$$

As the gear wheels G_1 and G_2 are connected, from the condition of compatibility, we can write

$$\omega_1 R_1 = \omega_2 R_2 = \frac{at}{r} R_2$$

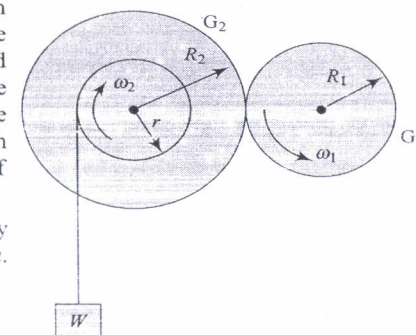


Fig. E9.18

$$\text{or } \omega_1 = \frac{at}{r} \frac{R_2}{R_1} \quad (3)$$

If θ_1 is the rotation of G_1 in time interval t ,

$$\omega_1 = \frac{d\theta_1}{dt} = \frac{at}{r} \frac{R_2}{R_1} \quad (4)$$

Integrating the above equation within limits, we obtain

$$\int_0^{\theta_1} d\theta_1 = \frac{a}{r} \frac{R_2}{R_1} \int_0^t t dt$$

$$\text{or } \theta_1 = \frac{at^2}{2r} \frac{R_2}{R_1} \quad (5)$$

Example 9.19 A pulley system attached with loads P and Q is shown in Fig. E9.19. If P has uniform acceleration of 4.2 m/s^2 with initial velocity 3 m/s downward, determine the number of revolutions executed by the pulley in 5 s , velocity and position of load Q after 5 s and acceleration of point A at $t = 0$. Take $r_P = 2 \text{ m}$, $r_Q = 1.5 \text{ m}$.

Solution Initial linear velocity of point A = 3 m/s . Therefore, initial angular velocity of pulley, $\omega_0 = \frac{3}{r_P} = 1.5 \text{ rad/s}$. Linear acceleration of point A = 4.2 m/s^2 . Therefore, angular acceleration of pulley,

$$\alpha = \frac{4.2}{2} = 2.1 \text{ rad/s}^2$$

Now, total angle rotated by the pulley in 5 s ,

$$\begin{aligned} \theta &= \omega_0 t + \frac{1}{2} \alpha t^2 = (1.5 \times 5) + \left(\frac{1}{2} \times 2.1 \times 5^2 \right) \\ &= 33.75 \text{ rad} \\ &= \frac{33.75}{2\pi} \approx 5.371 \text{ revolutions} \end{aligned}$$

Angular velocity of the pulley after 5 s ,

$$\omega = \omega_0 + \alpha t = 1.5 + 2.1 \times 5 = 12 \text{ rad/s}$$

So, linear velocity of Q = $\omega \times r_Q = 12 \times 1.5 = 18 \text{ m/s}$

Upward linear displacement of Q = $\theta \times r_Q = 33.75 \times 1.5 = 50.625 \text{ m}$.

Acceleration of point A will be the resultant of tangential and radial acceleration.

Here, $a_t = 4.2 \text{ m/s}^2$

Radial acceleration, $a_r = \omega_0^2 \times r_P = 1.5^2 \times 2 = 4.5 \text{ m/s}^2$

$$\begin{aligned} \text{Acceleration of point A, } a_A &= \sqrt{a_t^2 + a_r^2} \\ &= \sqrt{4.2^2 + 4.5^2} = 6.155 \text{ m/s}^2 \end{aligned}$$

Example 9.20 After making correction due to magnetic deviation and variation, an aeroplane's true heading is computed as 63° with air speed 295 kmph . Some checkpoints indicate that the plane is making a true course of 58° at a ground speed 250 kmph [Fig. E9.20(a)]. Determine the direction and the magnitude of the blowing wind.

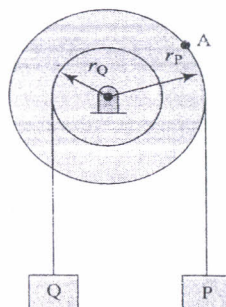
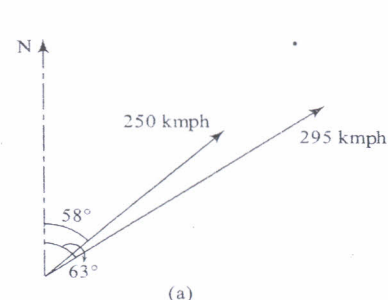
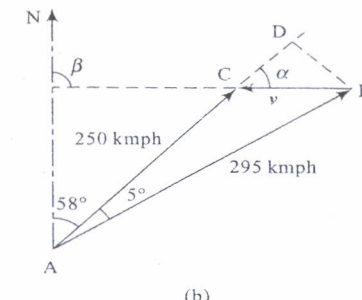


Fig. E9.19



(a)



(b)

Fig. E9.20

Solution From vector diagram, it can be said that velocity of wind $v = \overline{AB} - \overline{AC}$.

From geometry [Fig. E9.20(b)],

$$\sin 5^\circ = \frac{v \sin \alpha}{295}$$

$$\text{or } v \sin \alpha = 25.7109 \quad (1)$$

$$\text{Again, } \cos 5^\circ = \frac{250 + v \cos \alpha}{295}$$

$$\text{or } 250 + v \cos \alpha = 293.877$$

$$\text{or } v \cos \alpha = 43.877 \quad (2)$$

Now, from Eqs (1) and (2)

$$\tan \alpha = 0.585976$$

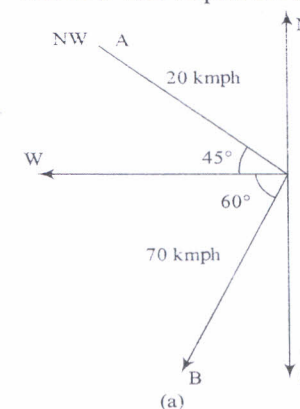
$$\text{or } \alpha = 30.369^\circ$$

Substituting in Eq. (1),

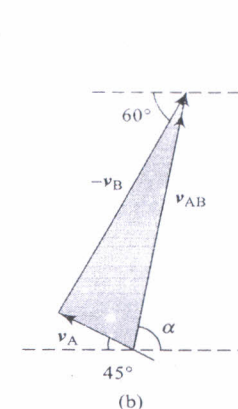
$$v = \frac{25.7109}{\sin 30.369^\circ} = 50.855 \text{ kmph}$$

Angle of wind velocity with respect to true north is $\beta = \alpha + 58 = 88.369^\circ$ towards left.

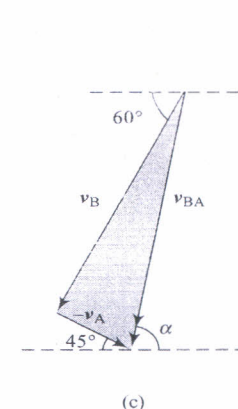
Example 9.21 A car A travels along a straight road heading northwest at a speed of 20 kmph . Another car B moves along a straight road 60° south of west at a speed of 70 kmph [Fig. E9.21(a)]. Determine the relative velocity of A with respect to B and that of B with respect to A.



(a)



(b)



(c)

Fig. E9.21

Solution The equation of speed equilibrium in the first case,

$$v_A = v_{AB} + v_B$$

or $v_{AB} = v_A - v_B$

This is represented by the vector diagram [Fig E9.21(b)]

Horizontal component of $v_{AB} = (-20 \cos 45^\circ + 70 \cos 60^\circ)$ kmph [towards right]

Vertical component of $v_{AB} = (20 \sin 45^\circ + 70 \sin 60^\circ)$ kmph [upwards]

So, the magnitude of v_{AB}

$$= \sqrt{(-20 \cos 45^\circ + 70 \cos 60^\circ)^2 + (20 \sin 45^\circ + 70 \sin 60^\circ)^2}$$

$$= 77.6188 \text{ kmph}$$

Inclination of this relative velocity,

$$\alpha = \tan^{-1} \left(\frac{20 \sin 45^\circ + 70 \sin 60^\circ}{-20 \cos 45^\circ + 70 \cos 60^\circ} \right) = 74.41^\circ \text{ in the north-east direction.}$$

To determine the relative velocity of B with respect to A, [Fig E9.21(c)]

$$v_B = v_{BA} + v_A$$

or $v_{BA} = v_B - v_A$

Observing the vector diagram in both the cases, we conclude that both the relative velocities have the same magnitude but act mutually opposite, i.e., along opposite directions, and with same inclination with the horizontal.

Example 9.22 A bar of length l making an angle θ with the vertical, moves its footpoint Q towards the right at constant speed V_Q [Fig. E9.22(a)]. Determine $\dot{\theta}$ and $\dot{\theta}$ in terms of V_Q , l , and θ .

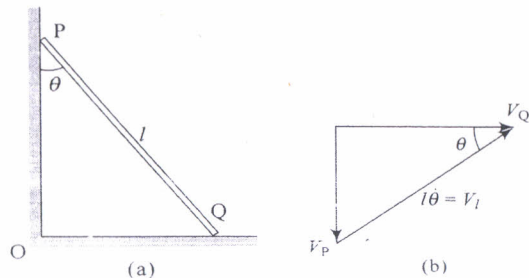


Fig. E9.22

Solution As Q moves towards the right, point P comes down, let us say, at speed V_P . At the same time, the rotational velocity of bar $\dot{\theta} = \omega$ and its linear projection, $V_l = l\dot{\theta}$. Let at certain time instant, the coordinates of P and Q be $(0, y_P)$ and $(x_Q, 0)$, respectively.

So, $V_P = \dot{y}_P$ and $V_Q = \dot{x}_Q$

From velocity triangle [Fig. E9.25(b)] $\cos \theta = \frac{V_Q}{l\dot{\theta}}$

or $\dot{\theta} = \frac{V_Q}{l \cos \theta}$ (1)

Taking the time derivative of Eq. (1), we obtain

$$\ddot{\theta} = \frac{V_Q}{l} \frac{d}{d\theta} \left(\frac{1}{\cos \theta} \right) \frac{d\theta}{dt} = \frac{V_Q}{l} \left(\frac{\sin \theta}{\cos^2 \theta} \right) \frac{V_Q}{l \cos \theta}$$

[Substituting from Eq. (1)]

$$= \frac{V_Q^2}{l^2} \frac{\tan \theta}{\cos^2 \theta}$$

Example 9.23 A roller of radius 2.5 m rolls without slipping along a horizontal plane as shown in Fig. 9.23(a). The centre O has uniform velocity 25 m/s. Determine the velocity of the points A and B of the roller.

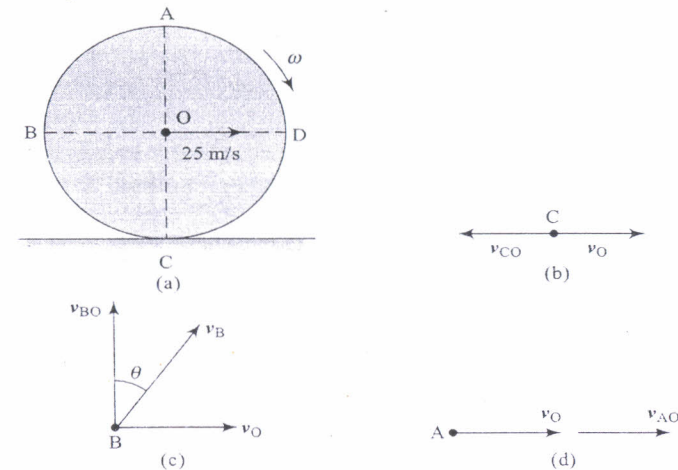


Fig. E9.23

Solution For no-slip rolling, velocity at point C with respect to centre O can be analysed as [Fig. E9.23(b)]

$$v_C = v_{CO} + v_O$$

$$0 = -r\omega + v_O$$

or $v_O = r\omega$

Considering magnitude, $\omega = 25/2.5 = 10 \text{ rad/s}$. This angular velocity is constant. In each case, the relative velocity of any point on the circumference of the roller remains constant and acts tangentially. Hence,

$$v_{BO} = v_{AO} = v_{DO} = \dots = r\omega$$

Analysing motion at point B [Fig. E9.23(c)], $v_B = v_{BO} + v_O$

$$\text{So, } v_B = \sqrt{(r\omega)^2 + 25^2} = \sqrt{(10 \times 2.5)^2 + 25^2} = 35.3553 \text{ m/s}$$

and inclination with vertical, $\theta = \tan^{-1}(25/25) = 45^\circ$. Similarly at point A [Fig. E9.26(d)],

$$v_A = v_{AO} + v_O$$

So, $v_A = 2 \times 25 = 50 \text{ m/s}$

Example 9.24 A roller moves within two conveyor belts, of which top belt moves at the speed of 8 m/s and bottom belt moves at 6 m/s. Determine the linear and angular velocity of the roller if the diameter of the roller is 55 cm [Fig. E9.24(a)].

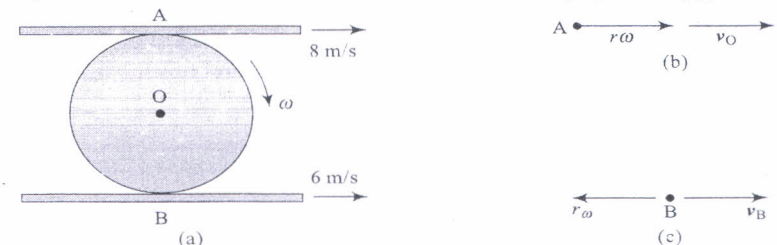


Fig. E9.24

Solution 1st approach Let the linear and angular velocity of centre O be v_O and ω , respectively, and velocities of top and bottom plates are v_P and v_B , respectively. Velocities of each point on the circumference, with respect to centre O is constant ($= r\omega$) acting tangentially. Now, velocity at point A [Fig. E9.24(b)],

$$v_A = v_{AO} + v_O$$

Hence, $8 = r\omega + v_O$

Similarly velocity at point B [Fig. E9.24(c)]

$$v_B = v_{BO} + v_O$$

Hence, $6 = -r\omega + v_O$

Solving Eqs (1) and (2) and substituting $r = (0.55/2)$ m, we obtain,

$$v_O = 7 \text{ m/s and } \omega = 3.636 \text{ rad/s}$$

2nd approach

Diameter of the roller = 55 cm

Let $r = \overline{BA} = 55j$, velocity at A, $v_A = 800i$, velocity at B, $v_B = 600i$, and $\bar{\omega}$ = angular velocity of roller = ωk

Now, $v_A = v_B + \bar{\omega} \times r$

$$\text{or } 800i = 600i + \omega k \times 55j$$

$$\text{or } 200i = -55\omega k$$

$$\text{or } \omega = -3.636$$

So, angular velocity of roller = 3.636 rad/s (clockwise).

$$\begin{aligned} \text{Again, } v_O &= v_B - 3.636k \times \frac{55}{2}j \\ &= 600i + 100i = 700i \end{aligned}$$

So, linear velocity of roller = 700 cm/s = 7 m/s.

3rd approach [Fig. E9.24(d)]

The instantaneous centre of velocity of AB will lie on AB since it is perpendicular to both v_A and v_B .

Let this centre be I. Then

$$\omega = \frac{v_A}{IA} = \frac{v_B}{IB} = \frac{v_O}{IO}$$

As v_A and v_B act along the same direction, IA and IB will also be of same sign. Hence the location of I will be outside of roller and on the extended AOB line. It is shown in Fig. E9.24(d).

$$\text{Now, } \omega = \frac{800}{IA} = \frac{600}{IB} = \frac{v_O}{IO}$$

$$\text{or } \omega = \frac{800 - 600}{IA - IB} = \frac{200}{55} = 3.636 \text{ rad/s}$$

$$\text{Therefore, } \frac{800}{IA} = \frac{600}{IB} = \frac{200}{55}$$

$$\text{So, } IA = \frac{800 \times 55}{200} = 220$$

$$IB = \frac{600 \times 55}{200} = 165$$

$$\text{Hence, } IC = IA - \frac{55}{2} = 192.5$$

$$\text{Again, } \frac{v_O}{IO} = \frac{200}{55}$$

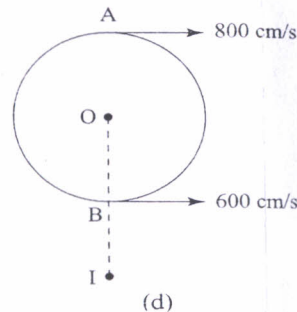


Fig. E9.24

$$\text{So, } v_O = \frac{200 \times 192.5}{55} = 700$$

Hence, linear velocity of roller = 700 cm/s = 7 m/s

Example 9.25 An epicyclic gear is shown in Fig. E9.25. The arm AB rotates clockwise at the rate of 70 rpm centred at B of the fixed large wheel. The fixed wheel rotates counterclockwise at the rate of 35 rpm. Determine the angular velocity of the small wheel.

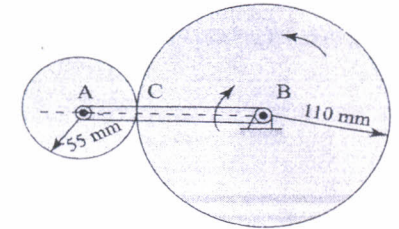


Fig. E9.25

Solution For arm AB

$$\overline{AB} = r_1 = 55i + 110i = 165i$$

$$\text{and } \bar{\omega}_1 = -70k$$

$$\begin{aligned} \text{Now, } v_A &= v_B + \bar{\omega}_1 \times r_1 = 0 - 70k \times 165i \\ &= -11550j \end{aligned}$$

Here C is the contact point of two wheels. Consider C as the point lying on the fixed wheel.

$$\text{So, } \overline{CB} = r_2 = 110i \text{ and } \bar{\omega} = 35k$$

$$\text{Now, } v_C = v_B + \bar{\omega} \times r_2 = 0 + 35k \times 110i = 3850j$$

Again consider C as the point lying on the small wheel.

$$\text{So, } \overline{AC} = r_o = 55i \text{ and } \bar{\omega}_o = \text{angular velocity of AC} = \omega k$$

$$\text{Now, } v_A = v_C + \bar{\omega}_o \times r_o$$

$$\text{or } -11550j = 3850j + \omega k \times 55i$$

$$\text{or } -15400j = 55\omega j$$

$$\text{or } \omega = -280.$$

So, angular velocity of the small wheel = 280 rpm clockwise.

Example 9.26 In the mechanism shown in Fig. E9.26, B is a hinge joint and end C of member BC moves on a horizontal surface. Determine (i) the velocity and acceleration of C, and (ii) angular velocity and angular acceleration of member BC. The member AB rotates clockwise about A at the rate of 3 rad/s.

Solution For member AB, angular velocity $\bar{\omega}_{AB} = -3k$.

$$\overline{AB} = 2i + 3j = r_{AB}$$

$$\text{Now, linear velocity } v_B = v_A + \bar{\omega}_{AB} \times r_{AB}$$

$$= 0 + (-3k) \times (2i + 3j) = -6j + 9i$$

$$\text{Angular velocity } \bar{\omega}_{BC} = \omega_{BC}k$$

$$\overline{BC} = 4i - 3j = r_{BC}$$

$$\text{Now, linear velocity } v_C = v_B + \bar{\omega}_{BC} \times r_{BC}$$

$$= (9i - 6j) + \omega_{BC}k \times (4i - 3j)$$

$$\text{or } v_C = 9i - 6j + 4\omega_{BC}j + 3\omega_{BC}i$$

$$= (9 + 3\omega_{BC})i + (4\omega_{BC} - 6)j$$

From the similarity theorem of algebraic equation, we can write

$$v_C = 9 + 3\omega_{BC}$$

$$\text{and } 4\omega_{BC} - 6 = 0$$

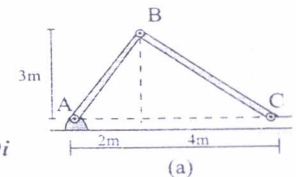


Fig. E9.26

Solving these two equations, we obtain, $\omega_{BC} = 1.5$ and $v_c = 13.5$.

So, velocity of C, $v_c = 13.5$ m/s and angular velocity of member BC, $\omega_{BC} = 1.5$ rad/s. Now we will find the corresponding acceleration.

For member AB

Angular acceleration $\bar{\alpha}_{AB} = 0$

Now, linear acceleration $a_B = a_A + \bar{\alpha}_{AB} \times r_{AB} + \bar{\omega}_{AB} \times (\bar{\omega}_{AB} \times r_{AB})$
 $= 0 + 0 - 3\mathbf{k} \times [-3\mathbf{k} \times (2\mathbf{i} + 3\mathbf{j})]$
 $= -3\mathbf{k} \times (-6\mathbf{j} + 9\mathbf{i}) = -18\mathbf{i} - 27\mathbf{j}$

For member BC

Assuming angular acceleration $\bar{\alpha}_{BC} = \alpha_{BC}\mathbf{k}$, linear acceleration is expressed as

$$a_C = a_B + \bar{\alpha}_{BC} \times r_{BC} + \bar{\omega}_{BC} \times (\bar{\omega}_{BC} \times r_{BC})$$

or, $a_C\mathbf{i} = (-18\mathbf{i} - 27\mathbf{j}) + \alpha_{BC}\mathbf{k} \times (4\mathbf{i} - 3\mathbf{j}) + 1.5\mathbf{k} \times [1.5\mathbf{k} \times (4\mathbf{i} - 3\mathbf{j})]$
 $= -(18\mathbf{i} + 27\mathbf{j}) + \alpha_{BC}\mathbf{k} \times (4\mathbf{i} - 3\mathbf{j}) + 1.5\mathbf{k} \times (6\mathbf{j} + 4.5\mathbf{i})$
 $= -(18\mathbf{i} + 27\mathbf{j}) + (4\alpha_{BC}\mathbf{j} + 3\alpha_{BC}\mathbf{i}) + (-9\mathbf{i} + 6.75\mathbf{j})$
 $= (-27 + 3\alpha_{BC})\mathbf{i} + (4\alpha_{BC} - 20.25)\mathbf{j}$

From the similarity theorem of algebraic equation, we obtain

$$a_C = 3\alpha_{BC} - 27 \quad (3)$$

$$0 = 4\alpha_{BC} - 20.25 \quad (4)$$

Solving the above equations, we obtain, $\alpha_{BC} = 5.0625$ and $a_C = -11.8125$. So, acceleration of C, $a_C = -11.8125$ m/s² and angular velocity of member BC, $\alpha_{BC} = 5.0625$ rad/s².

Example 9.27 A thick metal disk rolls without slip on a horizontal surface at the rate of 2 m/s towards left, as shown in Fig. E9.27. At a specific instant of motion, depicted by the instantaneous distances as shown, determine the angular velocity of the disk. The link MNP is attached with the disk at point M. Hence determine the angular velocities of members MN and NP.

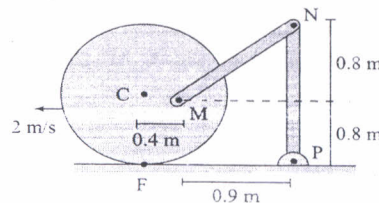


Fig. E9.27

Solution If the contact point F is considered as the instantaneous centre of velocity, then $v_F = 0$.

The velocity of centre C of the disk $v_C = -2\mathbf{i}$.

Now the radius vector $\overline{FC} = 0.8\mathbf{j} = r_{FC}$.

If the angular velocity of the disk is assumed as $\bar{\omega}$, we can write

$$v_C = v_F + \bar{\omega} \times r_{FC}$$

$$\text{or } -2\mathbf{i} = 0 + \omega\mathbf{k} \times 0.8\mathbf{j} = -0.8\omega\mathbf{i}$$

$$\text{or } \omega = \frac{-2}{-0.8} = 2.5$$

So angular velocity of disk = 2.5 rad/s.

Now radius vector $\overline{CM} = 0.4\mathbf{i} = r_{CM}$

Again we can write, $v_C = v_M + \bar{\omega} \times r_{CM}$

$$\text{or } -2\mathbf{i} = v_M + 2.5\mathbf{k} \times 0.4\mathbf{i} = v_M + \mathbf{j}$$

$$\text{or } v_M = -2\mathbf{i} - \mathbf{j}$$

For the member MN, angular velocity is $\bar{\omega}_{MN}$ and $r_{MN} = 0.9\mathbf{i} + 0.8\mathbf{j}$

$$\begin{aligned} \text{So, } v_N &= v_M + \bar{\omega}_{MN} \times r_{MN} = (-2\mathbf{i} - \mathbf{j}) + \omega_{MN}\mathbf{k} \times (0.9\mathbf{i} + 0.8\mathbf{j}) \\ &= (-2\mathbf{i} - \mathbf{j}) + 0.9\omega_{MN}\mathbf{j} - 0.8\omega_{MN}\mathbf{i} = (-2 - 0.8\omega_{MN})\mathbf{i} + (0.9\omega_{MN} - 1)\mathbf{j} \quad (1) \end{aligned}$$

For the member PN, angular velocity is $\bar{\omega}_{PN}$ and $r_{PN} = 1.6\mathbf{j}$

$$\begin{aligned} \text{So, } v_N &= v_P + \bar{\omega}_{PN} \times r_{PN} \\ &= 0 + \omega_{PN}\mathbf{k} \times 1.6\mathbf{j} = 1.6\omega_{PN}\mathbf{i} \quad (2) \end{aligned}$$

From (1) and (2) we obtain $(-2 - 0.8\omega_{MN})\mathbf{i} + (0.9\omega_{MN} - 1)\mathbf{j} = 1.6\omega_{PN}\mathbf{i}$

Hence from the above equation, we obtain

$$-2 - 0.8\omega_{MN} = 1.6\omega_{PN}$$

$$0.9\omega_{MN} - 1 = 0$$

Solving the above equations, we have $\omega_{MN} = 1.112$ and $\omega_{PN} = 1.806$. So, angular velocity of members MN and NP are 1.112 rad/s and 1.806 rad/s, respectively.

Example 9.28 At a particular instant of movement, as shown in Fig. E9.28, compute the velocity of N. Also determine the angular velocity of members PN and MN.

Solution For member QP

$v_Q = 0$ and angular velocity $\bar{\omega}_{QP} = -1.2\mathbf{k}$.

Radius vector $r_{QP} = \overline{QP} = 0.3\mathbf{j}$

Now, from the concept of relative velocity $v_P = v_Q + \bar{\omega}_{QP} \times r_{QP}$

$$\begin{aligned} &= 0 - 1.2\mathbf{k} \times 0.3\mathbf{j} \\ &= 0.36\mathbf{i} \end{aligned}$$

For member PN

Angular velocity of PN = $\bar{\omega}_{PN}$ and radius vector $r_{PN} = -0.5\mathbf{i} - 0.3\mathbf{j}$.

$$\begin{aligned} \text{Now, } v_N &= v_P + \bar{\omega}_{PN} \times r_{PN} = 0.36\mathbf{i} + \omega_{PN}\mathbf{k} \times (-0.5\mathbf{i} - 0.3\mathbf{j}) \\ &= 0.36\mathbf{i} - 0.5\omega_{PN}\mathbf{j} + 0.3\omega_{PN}\mathbf{i} = (0.36 + 0.3\omega_{PN})\mathbf{i} - 0.5\omega_{PN}\mathbf{j} \quad (1) \end{aligned}$$

For member MN

$v_M = 0$, angular velocity of MN = $\bar{\omega}_{MN}$ and radius vector $r_{MN} = -0.6\mathbf{j}$.

$$\begin{aligned} \text{Now, } v_N &= v_M + \bar{\omega}_{MN} \times r_{MN} \\ &= 0 + \omega_{MN}\mathbf{k} \times (-0.6\mathbf{j}) = 0.6\omega_{MN}\mathbf{i} \quad (2) \end{aligned}$$

From Eqs (1) and (2), we obtain

$$(0.36 + 0.3\omega_{PN})\mathbf{i} - 0.5\omega_{PN}\mathbf{j} = 0.6\omega_{MN}\mathbf{i}$$

$$\text{or } (0.36 + 0.3\omega_{PN} - 0.6\omega_{MN})\mathbf{i} - 0.5\omega_{PN}\mathbf{j} = 0$$

From similarity of algebraic equation, we obtain

$$\begin{aligned} 0.36 + 0.3\omega_{PN} - 0.6\omega_{MN} &= 0 \\ 0.5\omega_{PN} &= 0 \end{aligned}$$

Solving the above equations, we have, $\omega_{PN} = 0$ and $\omega_{MN} = 0.6$.

So, the linear velocity of N is 0.36 m/s horizontal rightwards, and the angular velocity of members PN and MN are zero and 0.6 rad/s respectively.

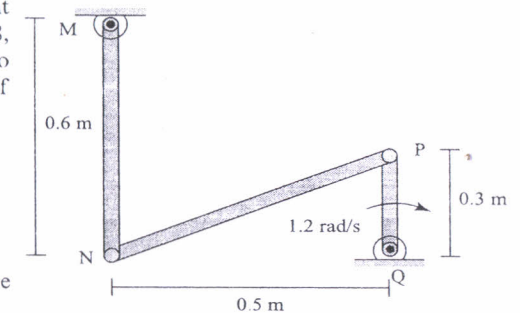


Fig. E9.28

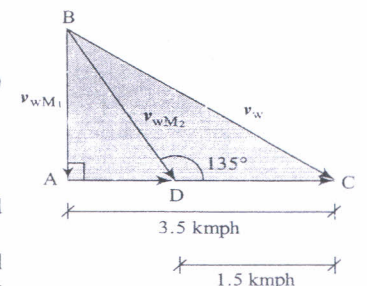


Fig. E9.29

Example 9.29 A man is walking due east at 3.5 kmph while the wind appears to be coming from north. If the walking speed is reduced to 1.5 kmph along the previous direction, the wind appears to be coming from northwest direction (Fig. E9.29). What is the speed of the wind?

Solution In the vector diagram BADC, BC represents the velocity of wind v_w , AC is the velocity of man in the first case, v_{M_1} , DC is the velocity of man in the second case, v_{M_2} . BA and BD represents the relative velocities v_{wM_1} and v_{wM_2} , respectively.

From the given condition, $\angle BAD = 90^\circ$ and $\angle BDC = 135^\circ$.

Using simple geometry, we find $\angle BDA = 180^\circ - 135^\circ = 45^\circ = \angle ABD$.

Now, $AD = 3.5 - 1.5 = 2$ kmph and hence $v_{wM_1} = 2$ kmph. Here,

$$v_w = \sqrt{v_{wM_1}^2 + v_{M_1}^2} = \sqrt{2^2 + 3.5^2} = 4.031 \text{ kmph}$$

Example 9.30 A bicycle wheel of 0.89 m diameter rolls without slip at 18 rad/s angular velocity [Fig. E9.30(a)]. If the angular acceleration becomes 13 rad/s², determine the velocity and the acceleration of point A on the tyre and B on the rim. The diameter of the rim is 0.78 m.

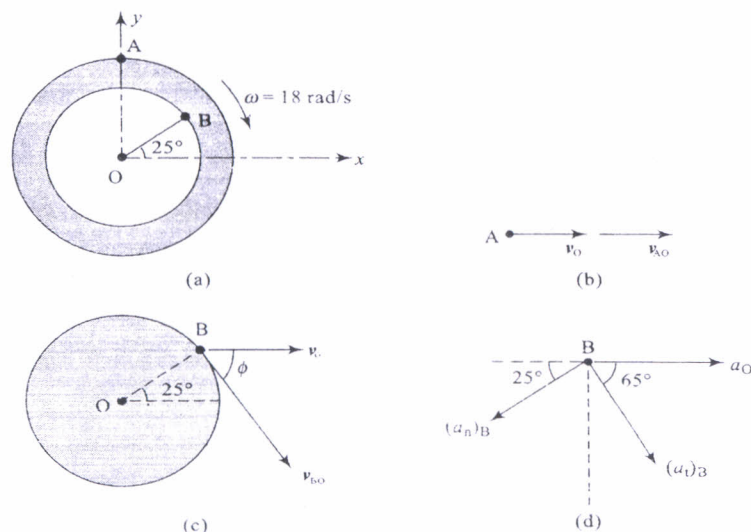


Fig. E9.30

Solution Linear velocity of centre point O [Fig. E9.30(b)],

$$v_O = \frac{0.89}{2} \times 18 = 8.01 \text{ m/s}$$

Relative velocity of point A with respect to centre O,

$$v_{AO} = r\omega = \frac{0.89}{2} \times 18 = 8.01 \text{ m/s}$$

So, velocity at A,

$$v_A = v_{AO} + v_O = 2 \times 8.01 = 16.02 \text{ m/s}$$

$$\text{Relative acceleration of A, } a_{AO} = r\alpha = \frac{0.89}{2} \times 13 = 5.785 \text{ m/s}^2$$

$$\text{Hence, tangential acceleration of A, } (a_t)_A = 5.785 \text{ m/s}^2$$

$$\text{Now, normal acceleration of A, } (a_n)_A = \frac{(v_{AO})^2}{r} = \frac{8.01^2}{0.89/2} = 144.18 \text{ m/s}^2$$

$$\text{So, the resultant acceleration of A} = \sqrt{(2 \times 5.785)^2 + 144.18^2} = 144.643 \text{ m/s}^2$$

$$\text{and inclination with the horizontal, } \theta = \tan^{-1} \left(\frac{144.18}{5.785} \right) = 87.7^\circ$$

Relative velocity of B with respect to centre O, acting tangentially [Fig. E9.30(c)]

$$v_{BO} = \frac{0.78}{2} \times 18 = 7.02 \text{ m/s}$$

Here the angle $\phi = 180^\circ - 2^\circ - 90^\circ = 65^\circ$

$$\text{So, } v_B = v_{BO} + v_O$$

$$\text{and } v_B = \sqrt{7.02^2 + 8.01^2 + 2 \times 7.02 \times 8.01 \cos 65^\circ} = 12.687 \text{ m/s}$$

$$\text{and inclination} = \sin^{-1} \left(\frac{7.02 \sin \phi}{12.687} \right) = 30.098^\circ \text{ with the horizontal}$$

$$\text{Tangential acceleration of B, } (a_t)_B = \frac{0.78}{2} \times 13 = 5.07 \text{ m/s}^2$$

$$\text{Normal acceleration of B, } (a_n)_B = \left(\frac{v_{BO}}{0.78/2} \right)^2 = \frac{(7.02)^2}{0.78/2} = 126.36 \text{ m/s}^2$$

Now, the acceleration at centre O, $a_O = (a_t)_A = 5.785 \text{ m/s}^2$.

At point B, the components of acceleration are shown. Resolving horizontally and vertically, $(a_B)_H = a_O + (a_t)_B \cos 65^\circ - (a_n)_B \cos 25^\circ$

$$= 5.785 + 5.07 \cos 65^\circ - 126.36 \cos 25^\circ$$

$$= -106.593 \text{ m/s}^2 = 106.593 \text{ m/s}^2 \text{ (towards left)}$$

$$\text{and } (a_B)_V = (a_t)_B \sin 65^\circ + (a_n)_B \sin 25^\circ = 57.997 \text{ m/s}^2 \text{ (downwards)}$$

$$\text{So, finally acceleration of point B} = \sqrt{(-106.593)^2 + (57.997)^2} = 121.35 \text{ m/s}^2$$

$$\text{and inclined } \tan^{-1} \left(\frac{57.997}{106.593} \right) = 28.55^\circ \text{ with the horizontal.}$$

Example 9.31 A double gear arrangement as shown in Fig. E9.31(a) rolls on a lower rack, which is stationary. The velocity of centre O is 2 m/s towards the right. Determine the angular velocity of the gear. Also determine the velocity of the upper rack and point B.

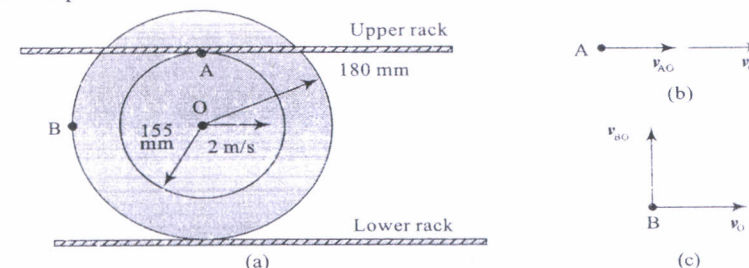


Fig. E9.31

Solution Let us assume that in time interval t , the gear rotates by an amount θ , thus making linear displacement S . So, $S = 0.18\theta$. Differentiating this with respect to time,

$$\frac{dS}{dt} = 0.18 \frac{d\theta}{dt}$$

or $v_O = 0.18\omega$

or $\omega = \frac{v_O}{0.18} = \frac{2}{0.18} = 11.112 \text{ rad/s}$

At points A and B, the relative velocities are

$$v_{AO} = 0.155\omega = 0.155 \times \frac{2}{0.18} = 1.722 \text{ m/s}$$

$$v_{BO} = 0.18\omega = 0.18 \times \frac{2}{0.18} = 2 \text{ m/s}$$

For point A [Fig. E9.31(b)], $v_A = v_{AO} + v_O$

Hence, $v_A = 2 \times 1.722 = 3.444 \text{ m/s}$

For point B [Fig. E9.31(c)], $v_B = v_{BO} + v_O$

Therefore,

$$v_B = \sqrt{2^2 + 2^2} = 2.828 \text{ m/s with inclination } 45^\circ \text{ with the horizontal.}$$

Example 9.32 A heavy prismatic bar AB is rotating at an angular speed of 6.2 rad/s. The mass centre C of the bar, thus moves upward at 115 m/s (Fig. E9.32). At this instant, determine the velocity of the topmost point A of the bar.

Solution Velocity of point C, $v_C = 115 \text{ m/s}$ upwards. Relative velocity of point A, with respect to C, acting tangentially

$$v_{AC} = 13\omega = 13 \times 6.2 = 80.6 \text{ m/s}$$

Now, velocity of point A,

$$v_A = v_{AC} + v_C$$

So, $v_A = \sqrt{80.6^2 + 115^2} = 140.43 \text{ m/s}$

and inclined with the horizontal by an angle

$$\tan^{-1} \left(\frac{115}{80.6} \right) = 54.97^\circ$$

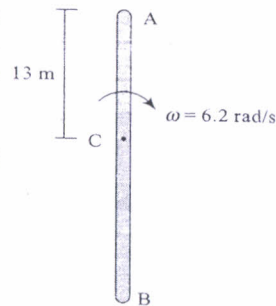


Fig. E9.32

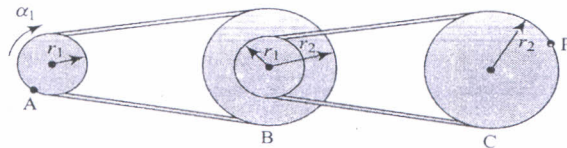


Fig. E9.33

Solution Considering same linear displacement of belt in pulley A and pulley B, angular displacements θ_1 and θ_B , and angular velocities ω_1 and ω_B , respectively, we can write,

$$\theta_B \times r_2 = \theta_1 \times r_1 \text{ or } \theta_B = \left(\frac{r_1}{r_2} \right) \theta_1$$

$$\text{and } \omega_B \times r_2 = \omega_1 \times r_1 \text{ or } \omega_B = \left(\frac{r_1}{r_2} \right) \omega_1$$

Now, if the pulley A attains angular speed ω_1 in time t , $\omega_1 = \alpha_1 t$

where α_1 is the constant acceleration. So, $\omega_B = \left(\frac{r_1}{r_2} \right) \alpha_1 t$

Again considering the condition of compatibility between internal pulley of B and pulley C, we obtain, $\omega_B \times r_1 = \omega_C \times r_2$

$$\text{or } \omega_C = \left(\frac{r_1}{r_2} \right) \omega_B = \left(\frac{r_1}{r_2} \right) \left(\frac{r_1}{r_2} \right) \alpha_1 t = \left(\frac{r_1}{r_2} \right)^2 \alpha_1 t$$

Following the same logic as mentioned above, we can write, $\theta_C \times r_2 = \theta_B \times r_1$

$$\text{or } \theta_C = \left(\frac{r_1}{r_2} \right) \theta_B = \left(\frac{r_1}{r_2} \right) \left(\frac{r_1}{r_2} \right) \theta_1 = \left(\frac{r_1}{r_2} \right)^2 \theta_1$$

Differentiating the above relation twice, we get

$$\ddot{\theta}_C = \ddot{\theta}_B = \left(\frac{r_1}{r_2} \right)^2 \ddot{\theta}_1 = \left(\frac{r_1}{r_2} \right)^2 \alpha_1$$

So, tangential acceleration at P, $(a_t)_P = \ddot{\theta}_P \times r_2 = \frac{r_1^2}{r_2^2} \alpha_1$

Normal acceleration at P, $(a_n)_P = \omega_C^2 \times r_2 = \left(\frac{r_1}{r_2} \right)^4 \alpha_1^2 t^2 r_2 = \frac{r_1^4}{r_2^3} \alpha_1^2 t^2$

Total acceleration at P,

$$a_P = \sqrt{(a_t)_P^2 + (a_n)_P^2} = \left[\frac{r_1^4}{r_2^2} \alpha_1^2 + \frac{r_1^8}{r_2^6} \alpha_1^4 t^4 \right]^{1/2}$$

$$= \left(\frac{r_1^4}{r_2^2} \alpha_1^2 \right)^{1/2} \left(1 + \frac{r_1^4}{r_2^4} \alpha_1^2 t^4 \right)^{1/2} = \frac{r_1^2}{r_2} \alpha_1 \left[1 + \left(\frac{r_1}{r_2} \right)^4 \alpha_1^2 t^4 \right]^{1/2}$$

Example 9.34 The slotted arm OA rotates with constant angular velocity $\omega = \dot{\theta}$ during a limited interval of its motion and moves the pivoted slider block along the horizontal slot (Fig. E9.34). Write the expressions for velocity v_B and acceleration a_B of the pin B in the slider block in terms of θ .

Solution When the arm rotates by an amount θ , horizontal displacement is x , due to velocity v_B . Now from simple geometry,

$$x = a \tan \theta \quad (1)$$

Differentiating Eq. (1) once, we obtain,

$$\dot{x} = a \sec^2 \theta \dot{\theta}$$

or $v_B = a \omega \sec^2 \theta \quad (2)$

Differentiating Eq. (2) once, we obtain,

$$a_B = \frac{d}{dt} (a \omega \sec^2 \theta)$$

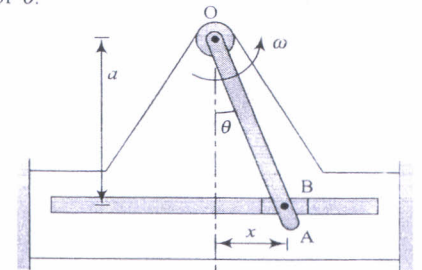


Fig. E9.34

$$\begin{aligned}
 &= \frac{d}{dt} [a\omega (1 + \tan^2 \theta)] \\
 &= a(2\tan \theta \sec^2 \theta) \times \dot{\theta} \times \omega + \dot{a} \omega (1 + \tan^2 \theta) \\
 &= 2a\omega^2 \tan \theta \sec^2 \theta \quad [\text{For constant angular velocity, } \dot{\omega} = 0]
 \end{aligned}$$

Example 9.35 Derive an expression for the angular velocity ω of the slender bar PQ as a function of distance x and the constant angular velocity ω_0 of the drum (Fig. E9.35).

Solution Now, linear velocity of the string,
 $\dot{x} = -\omega_0 r$

From simple geometry,

$$\tan \theta = \frac{h}{x} \quad (1)$$

Differentiating the above relation once, we get,

$$\sec^2 \theta \dot{\theta} = -\frac{h}{x^2} \dot{x}$$

$$\text{or} \quad (1 + \tan^2 \theta) \omega = -\frac{h}{x^2} (-r\omega_0)$$

$$\text{or} \quad \left(1 + \frac{h^2}{x^2}\right) \omega = -\frac{h}{x^2} (-r\omega_0)$$

$$\text{or} \quad (x^2 + h^2) \omega = rh\omega_0$$

$$\text{Hence, } \omega = \frac{rh\omega_0}{x^2 + h^2}$$

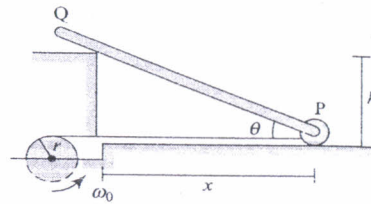


Fig. E9.35

Example 9.36 The hydraulic cylinder B gives end A of link AC a constant velocity v_0 in the negative x -direction (Fig. E9.36). Determine the expression for the angular velocity $\omega = \dot{\theta}$ and angular acceleration $\alpha = \ddot{\theta}$ of the link in terms of x .

Solution As per the given condition,

$$v_0 = -\dot{x}$$

Geometry of the arrangement suggests that,

$$\cos \theta = \frac{x}{l} \quad (1)$$

Differentiating Eq. (1) with respect to time, we get

$$-\sin \theta \dot{\theta} = \frac{1}{l} \dot{x}$$

Putting the values, we get

$$-(1 - \cos^2 \theta)^{1/2} \omega = \frac{1}{l} (-v_0)$$

$$\text{or} \quad \left(1 - \frac{x^2}{l^2}\right)^{1/2} \omega = \frac{v_0}{l}$$

$$\text{or} \quad \frac{\sqrt{l^2 - x^2}}{l} \omega = \frac{v_0}{l}$$

$$\text{Hence, } \omega = \frac{v_0}{\sqrt{l^2 - x^2}} \quad (2)$$

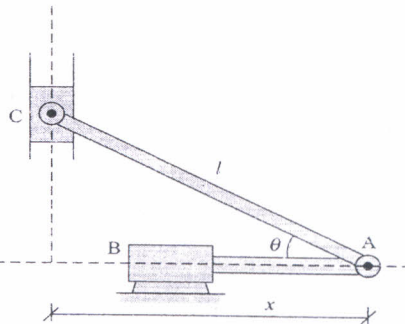


Fig. E9.36

Differentiating Eq. (2) with respect to time, we get

$$\dot{\omega} = -\frac{1}{2} (l^2 - x^2)^{-3/2} (-2x) \dot{x} v_0$$

or $\alpha = x v_0 (l^2 - x^2)^{-3/2} (-v_0)$ [Substituting $\dot{x} = -v_0$]

$$= \frac{-x v_0^2}{(l^2 - x^2)^{3/2}}$$

Example 9.37 In the quadratic crank mechanism as shown in Fig. E9.37, determine the angular velocity of AD and the velocity of point D for the phase indicated. Assume, $\omega_{BC} = 10$ rad/s.

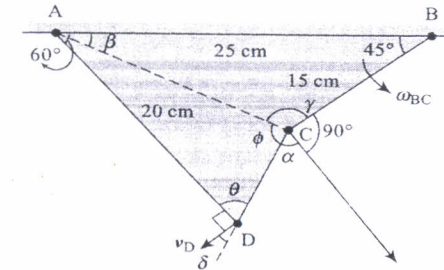


Fig. E9.37

Solution The velocity of point C on $\overline{BC}$, acting normal to $\overline{BC}$ at C,

$$v_C = \omega_{BC} \times \overline{BC} = 10 \times 0.15 = 1.5 \text{ m/s}$$

The component of v_C along CD towards D,

$$v_{CD} = v_C \cos \alpha \quad (1)$$

Now, consider the direction of v_D and inclination δ with the direction of v_{CD} . So,

$$v_D \cos \delta = v_{CD}$$

$$\text{or} \quad v_D = \frac{v_{CD}}{\cos \delta} = v_C \times \frac{\cos \alpha}{\cos \delta} = 1.5 \times \frac{\cos \alpha}{\cos \delta} \quad (2)$$

Angular velocity of link AD,

$$\begin{aligned}
 \omega_{AD} &= \frac{v_D}{AD} = \frac{v_C}{AD} \times \frac{\cos \alpha}{\cos \delta} \\
 &= \frac{1.5}{0.2} \times \frac{\cos \alpha}{\cos \delta} = 7.5 \frac{\cos \alpha}{\cos \delta} \quad (3)
 \end{aligned}$$

Applying geometry on $\triangle ABC$,

$$AC = [AB^2 + BC^2 - 2AB \times BC \cos 45^\circ]^{1/2} = 17.88 \text{ cm} = 0.1788 \text{ m}$$

Applying the law of sine on $\triangle ABC$, $\frac{AB}{\sin \gamma} = \frac{BC}{\sin \beta} = \frac{CA}{\sin 45^\circ}$

$$\text{or} \quad \gamma = \sin^{-1} \left[\frac{AB}{CA} \sin 45^\circ \right] = 81.37^\circ$$

$$\beta = \sin^{-1} \left[\frac{BC}{CA} \sin 45^\circ \right] = 36.39^\circ$$

$$\text{So, } \angle DAC = 60^\circ - \beta = 23.61^\circ$$

Now, from $\triangle DAC$,

$$DC = [AC^2 + AD^2 - 2AC \times AD \cos 23.61^\circ]^{1/2} = 8.02 \text{ cm} = 0.0802 \text{ m}$$

$$\text{Again, } \frac{AD}{\sin \phi} = \frac{DC}{\sin 23.61^\circ}$$

$$\text{or } \phi = \sin^{-1} \left[\frac{AD}{DC} \sin 23.61^\circ \right] = 87.16^\circ$$

Now, at point C,

$$\alpha = 360^\circ - (\gamma + \phi + 90^\circ) = 101.47^\circ$$

At point D,

$$\theta = 180^\circ - (23.61^\circ + \phi) = 69.23^\circ$$

Again at point D,

$$\delta = 180^\circ - (90^\circ + \theta) = 20.77^\circ$$

$$\text{From Eq. (2), } v_D = 1.5 \times \frac{\cos 101.47^\circ}{\cos 20.77^\circ} = -0.319 \text{ m/s}$$

From Eq. (3),

$$\omega_{AD} = 7.5 \times \frac{\cos 101.47^\circ}{\cos 20.77^\circ} = -1.595 \text{ rad/s} = 1.595 \text{ rad/s (clockwise)}$$

Example 9.38 In the four bar linkage, shown in Fig. E9.38(a), crank AB is rotating counterclockwise at 3 rad/s. Determine the linear velocities of points B and C and angular velocities of BC and DC.

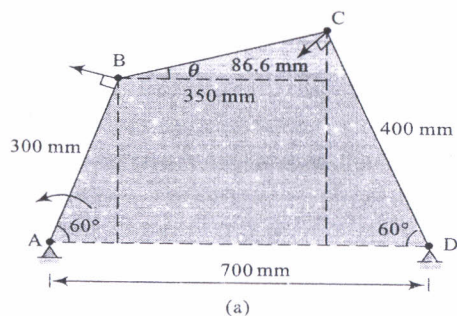
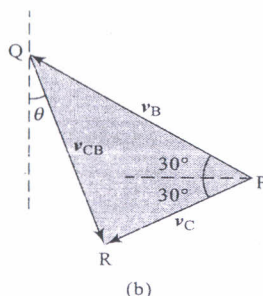


Fig. E9.38



Solution Here $v_B = \omega_{AB} \times AB = 3 \times 0.3 = 0.9 \text{ m/s}$ acts normal to AB at point B and directing left. Now, considering relative motion of C with respect to B,

$$v_C = v_{CB} + v_B$$

Considering geometry of the configuration [Fig. E9.38(b)],

$$BC = \sqrt{350^2 + 86.6^2} = 360.55 \text{ mm}$$

$$\text{and } \theta = \tan^{-1} \left(\frac{86.6}{350} \right) = 13.9^\circ$$

Here the velocity v_{CB} will act normal to link BC, that is, at an angle $\theta = 13.9^\circ$ with the vertical. Using law of sine in velocity triangle ΔPQR , we obtain

$$\frac{v_{CB}}{\sin 60^\circ} = \frac{v_C}{\sin(60^\circ - \theta)} = \frac{v_B}{\sin[180^\circ - 60^\circ - (60^\circ - \theta)]}$$

$$\text{or } \frac{v_{CB}}{\sin 60^\circ} = \frac{v_C}{\sin 46.1^\circ} = \frac{0.9}{\sin 73.9^\circ}$$

Therefore, $v_C = 0.675 \text{ m/s}$ and $v_{CB} = 0.811 \text{ m/s}$

$$\text{Now angular velocity of DC, } \omega_{DC} = \frac{v_C}{DC} = \frac{0.675}{0.4} = 1.6875 \text{ rad/s}$$

$$\text{Angular velocity of BC, } \omega_{BC} = \frac{v_{CB}}{BC} = \frac{0.811}{0.36055} = 2.249 \text{ rad/s}$$

Example 9.39 In the previous example, determine the linear accelerations of points B and C and angular accelerations of BC and CD.

Solution The acceleration of point C as a point on BC will be (Fig. E9.39),

$$a_C = (a_{CB})_t + (a_{CB})_n + a_B \quad (1)$$

Again C being a point on CD,

$$a_C = (a_{CD})_t + (a_{CD})_n$$

Substituting in Eq. (1),

$$(a_{CD})_t + (a_{CD})_n = (a_{CB})_t + (a_{CB})_n + a_B \quad (2)$$

$$\text{Now, } (a_B)_n = \frac{v_B^2}{AB} = \frac{900^2}{300} = 2700 \text{ mm/s}^2$$

$$(a_{CB})_n = \frac{v_{CB}^2}{CB} = \frac{811^2}{360.55} = 1824.22 \text{ mm/s}^2$$

$$(a_{CD})_n = \frac{v_C^2}{CD} = \frac{675^2}{400} = 1139.06 \text{ mm/s}^2$$

$$(a_{CD})_t = \alpha_{DC} \times 400$$

$$(a_{CB})_t = \alpha_{BC} \times 360.55$$

Substituting these parameters in Eq. (2) and resolving along x- and y-components, we obtain

$$400 \alpha_{DC} \cos 30^\circ + 1139.06 \cos 60^\circ = -360.55 \alpha_{BC} \sin 13.9^\circ - 1824.22 \cos 13.9^\circ - 2700 \cos 60^\circ$$

$$\text{and } 400 \alpha_{DC} \sin 30^\circ - 1139.06 \sin 60^\circ = 360.55 \alpha_{BC} \cos 13.9^\circ - 1824.22 \sin 13.9^\circ - 2700 \sin 60^\circ$$

$$\text{or } \begin{cases} 346.41 \alpha_{DC} + 86.61 \alpha_{BC} = -3690.33 \\ 200 \alpha_{DC} - 349.99 \alpha_{BC} = -1790.04 \end{cases}$$

Solving the above equations, we obtain, $\alpha_{DC} = -10.44$ and $\alpha_{BC} = -0.85$

So, angular acceleration of BC is 0.85 rad/s^2 counterclockwise and that of CD, 10.44 rad/s^2 clockwise.

Example 9.40 A road roller of 3 m diameter rolls to the right with $\omega = 8 \text{ rad/s}$ and $\alpha = 4 \text{ rad/s}^2$ as shown in Fig. E9.40(a). Determine the linear velocity and the acceleration at top point A on the wheel.

Solution Path traversed during time t ,

$$P_1 C_1' = R\theta = PC_1' \quad [\text{as point P is shifted to } P_1]$$

So, linear velocity at centre of the roller, $v_C = R\omega = 1.5 \times 8 = 12 \text{ m/s}$

Linear acceleration at centre of the roller, $a_C = R\alpha = 1.5 \times 4 = 6 \text{ m/s}^2$

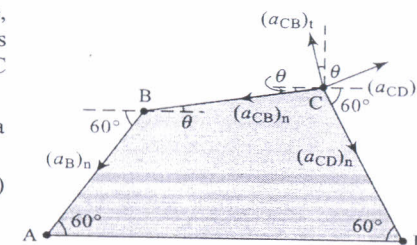


Fig. E9.39

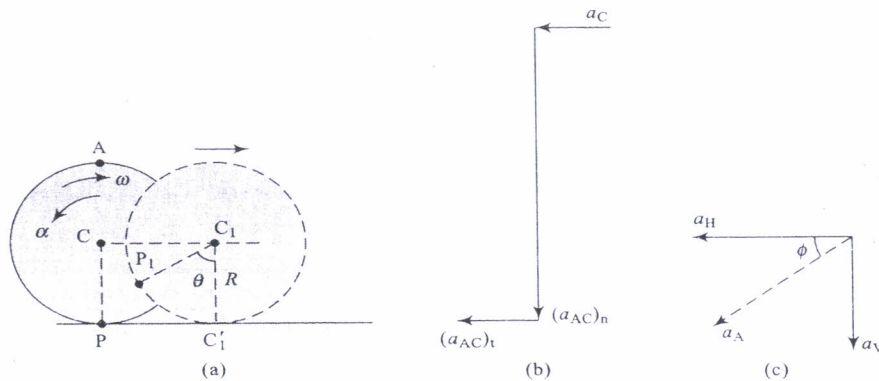


Fig. E9.40

Now, the velocity equation of point A

$$v_A = v_{AC} + v_C$$

Again, $v_{AC} = AC \times \omega = 1.5 \times 8 = 12 \text{ m/s}$

Here as v_{AC} and v_C both are horizontal to the right,

$$v_A = 2 \times 12 = 24 \text{ m/s}$$

Now the equation of acceleration of point A,

$$a_A = (a_{AC})_t + (a_{AC})_n + a_C$$

Here $(a_{AC})_t = R\alpha = 1.5 \times 4 = 6 \text{ m/s}^2$ (left)

$$(a_{AC})_n = \omega^2 R = 8^2 \times 1.5 = 96 \text{ m/s}^2 \text{ (downwards)}$$

The direction of all components of acceleration are shown in Figs E9.40 (b) and (c). Resolving all components along the horizontal and vertical directions,

$$a_H = a_C + (a_{AC})_t = 12 \text{ m/s}^2 \text{ (←)}$$

$$a_V = (a_{AC})_n = 96 \text{ (↓)}$$

$$\text{So, } a_A = \sqrt{12^2 + 96^2} = 96.747 \text{ m/s}^2$$

$$\text{and inclination with horizontal, } \phi = \tan^{-1} \left(\frac{96}{12} \right) = 82.87^\circ$$

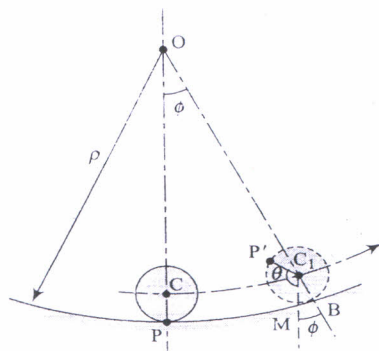


Fig. E9.41

Example 9.41 A roller of radius r rolls on the circular plane of radius ρ , as shown in Fig. E9.41. Study the motion.

Solution Initial position of roller is shown with centre C. When the point P moves to P', angle subtended at O is ϕ . Now, the arc length $BMP' = \text{arc length PB}$

$$\text{So, } r(\theta + \phi) = \rho\phi$$

$$\text{or } \theta = \frac{\rho - r}{r} \phi \quad (1)$$

Differentiating Eq. (1) with respect to time, once and twice, we obtain

$$\frac{d\theta}{dt} = \frac{\rho - r}{r} \frac{d\phi}{dt}$$

$$\text{and } \frac{d^2\theta}{dt^2} = \frac{\rho - r}{r} \frac{d^2\phi}{dt^2} \quad (2)$$

Now, if linear velocity of centre C is v_C and acceleration $(a_C)_t$ acting tangentially, the angular velocity and angular acceleration are expressed as,

$$\omega_C = \frac{d\phi}{dt} = \frac{v_C}{\rho - r} \quad \text{and} \quad \alpha_C = \frac{d^2\phi}{dt^2} = \frac{(a_C)_t}{\rho - r}$$

Substituting in Eq. (2), we obtain

$$\omega = \frac{d\theta}{dt} = \frac{\rho - r}{r} \frac{v_C}{\rho - r} = \frac{v_C}{r}$$

$$\text{and } \alpha = \frac{d^2\theta}{dt^2} = \frac{\rho - r}{r} \frac{(a_C)_t}{\rho - r} = \frac{(a_C)_t}{r}$$

Example 9.42 In the epicyclic gear train, as shown in Fig. E9.42(a), the arm is moving 6 rad/s clockwise having acceleration 10 rad/s² counterclockwise. Compute the linear velocity and the acceleration of point A in the phase shown in Fig. E9.42(a). Take help of the previous example.

Solution As done in the previous example, if ϕ is the angular displacement of arm OC, angular velocity ω , and angular acceleration α of the small wheel about its own centre,

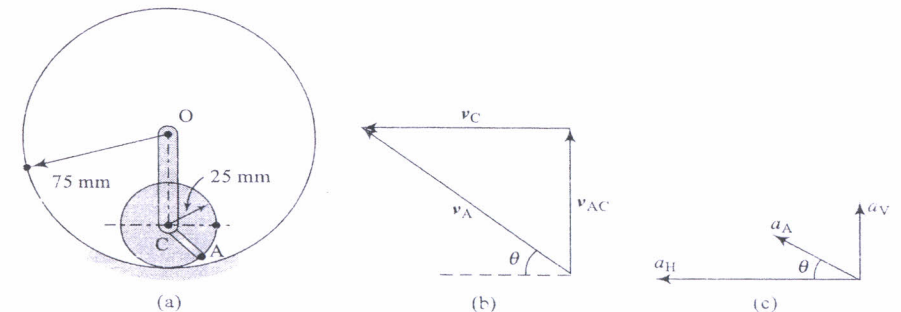


Fig. E9.42

$$\left. \begin{aligned} \omega &= \frac{\rho - r}{r} \frac{d\phi}{dt} \\ \alpha &= \frac{\rho - r}{r} \frac{d^2\phi}{dt^2} \end{aligned} \right\} \quad (1)$$

$$\text{Here, } \rho = 75 \text{ mm, } r = 25 \text{ mm, } \frac{d\phi}{dt} = 6 \text{ rad/s, } \frac{d^2\phi}{dt^2} = 10 \text{ rad/s}^2$$

Substituting in Eq. (1),

$$\left. \begin{aligned} \omega &= \frac{75 - 25}{25} \times 6 = 12 \text{ rad/s} \\ \alpha &= \frac{75 - 25}{25} \times 10 = 20 \text{ rad/s}^2 \end{aligned} \right\}$$

To study the motion of A [Fig. E9.42(b)],

$$v_A = v_{AC} + v_C$$

Here,

$$v_{AC} = \omega r = 12 \times 25 = 300 \text{ mm/s vertically upwards}$$

$$v_C = (\rho - r) \frac{d\phi}{dt} = \omega r = 300 \text{ mm/s horizontally left}$$

So, $v_A = \sqrt{300^2 + 300^2} = 424.26 \text{ mm/s}$ and inclined at 45° .

Again the equilibrium equation of acceleration of point A is [Fig. E9.42(c)]

$$a_A = (a_{AC})_t + (a_{AC})_n + (a_C)_t + (a_C)_n \quad (2)$$

Here, $(a_{AC})_t = \alpha r = 20 \times 25 = 500 \text{ mm/s}^2$ vertically downwards

$$(a_{AC})_n = \omega^2 r = 12^2 \times 25 = 3600 \text{ mm/s}^2 \text{ towards left}$$

$$(a_C)_t = (\rho - r) \times \frac{d^2\phi}{dt^2} = r\alpha = 500 \text{ mm/s}^2 \text{ towards right}$$

$$(a_C)_n = (\rho - r) \left(\frac{d\phi}{dt} \right)^2 = (75 - 25) \times 6^2 = 1800 \text{ mm/s}^2 \text{ vertically upwards}$$

All these components of Eq. (2) finally reduce to

$$a_H = 3600 - 500 = 3100 \text{ mm/s}^2 \text{ towards leftward}$$

$$a_V = 1800 - 500 = 1300 \text{ mm/s}^2 \text{ upwards}$$

Now, $a_A = \sqrt{3100^2 + 1300^2} = 3361.55 \text{ mm/s}^2$ and inclination with the horizontal = 22.75° .

Example 9.43 The rod as shown in Fig. E9.43(a) has a velocity 6 m/s vertically at point Q. Determine the angular speed of PQ, velocity of end P, and velocity of middle point R of PQ. The length of the rod is 2.1 m. Use the method of instantaneous centre.

Solution Figure E9.43(b) shows the direction of velocities at P, Q, and R and centre of instantaneous rotation I. About this point I, each point has same angular speed ω .

Now, $IQ = 2.1 \cos 30^\circ = 1.8187 \text{ m}$

$$IP = 2.1 \sin 30^\circ = 1.05 \text{ m} \quad \text{and} \quad RP = 2.1/2 = 1.05 \text{ m}$$

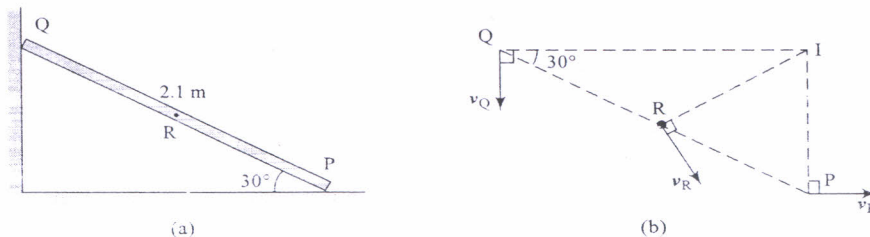


Fig. E9.43

Here, $v_Q = \omega \times IQ$

$$\text{or} \quad \omega = \frac{v_Q}{IQ} = \frac{6}{1.8187} = 3.299 \text{ rad/s}$$

$$\text{Now,} \quad v_P = \omega \times IP = 3.299 \times 1.05 = 3.464 \text{ m/s}$$

In $\triangle IRP$, $IP = IR$ and $\angle IPR = 60^\circ$. So $\triangle IRP$ is equilateral. Hence, $IR = 1.05 \text{ m}$.

$$\text{So,} \quad v_R = \omega \times IR = 3.299 \times 1.05 = 3.464 \text{ m/s}$$

Example 9.44 A slender prismatic bar PR supports at any point Q on the vertical wall [Fig. E9.44(a)]. If the point P has a velocity constant in value and directed horizontally, determine the angular speed of the bar.

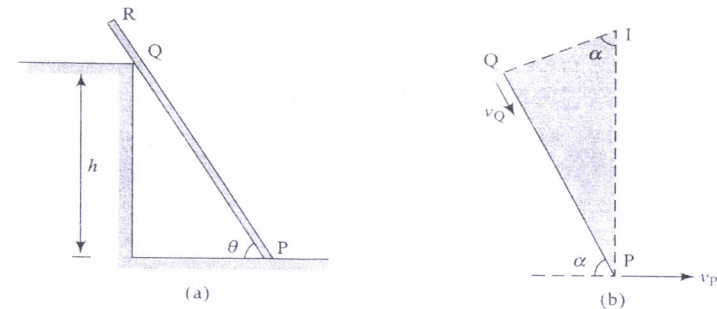


Fig. E9.44

Solution The velocity at support Q will be along the bar. The perpendiculars drawn on the lines of action of v_P and v_Q intersect at I, the instantaneous centre of rotation [Fig. E9.44(b)].

Using simple geometry, $PQ \sin \alpha = h$

$$\text{or} \quad PQ = h \operatorname{cosec} \alpha \quad (1)$$

Again from velocity diagram, $IP \sin \alpha = PQ$

$$\text{or} \quad IP = PQ \operatorname{cosec} \alpha = h \operatorname{cosec}^2 \alpha$$

[Substituting from Eq. (1)]

$$\text{Now,} \quad v_P = \omega \times IP$$

$$\text{or} \quad \omega = \frac{v_P}{IP} = \frac{v_P}{h} \sin^2 \alpha$$

Example 9.45 In the configuration of the solid bar PQ, shown in Fig. E9.45(a) with $v_P = 2.4 \text{ m/s}$ constant, determine the angular speed of PQ and the velocity of end Q.

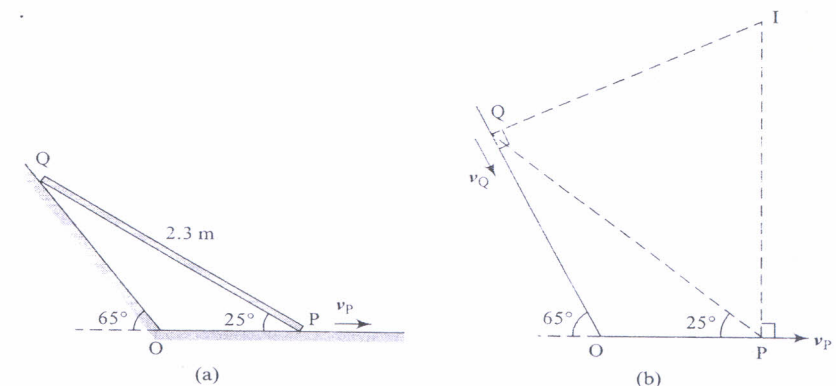


Fig. E9.45

Solution The velocity diagram with instantaneous centre of rotation is shown in Fig. E9.45(b). From geometry,

$$\angle OQP = 65^\circ - 25^\circ = 40^\circ$$

$$\text{So,} \quad \angle IQP = 90^\circ - 40^\circ = 50^\circ$$

$$\angle IPQ = 90^\circ - 25^\circ = 65^\circ$$

$$\text{and} \quad \angle PIQ = 180^\circ - (50^\circ + 65^\circ) = 65^\circ$$

Now, applying the rule of sine in ΔPIQ ,

$$\frac{PQ}{\sin \angle PIQ} = \frac{IP}{\sin \angle IQP} = \frac{IQ}{\sin \angle IPQ}$$

or $\frac{2.3}{\sin 65^\circ} = \frac{IP}{\sin 50^\circ} = \frac{IQ}{\sin 65^\circ}$

So, $IQ = 2.3 \text{ m}$ and $IP = 1.944 \text{ m}$

Again, $\omega \times IP = v_P$

or $\omega = \frac{v_P}{IP} = \frac{2.4}{1.944} = 1.235 \text{ rad/s}$

So, $v_Q = \omega \times IQ = 1.235 \times 2.3 = 2.84 \text{ m/s}$

Example 9.46 A slender bar PQ rests on a hemispherical disc of radius 5.3 cm [Fig. E9.46(a)]. If end P moves horizontally at 3.2 m/s, determine the angular speed of the bar.

Solution Draw the velocity diagram with instantaneous centre of rotation [Fig. E9.46(b)]. From geometry, $OQ = 5.3 \text{ cm}$. From ΔPOQ ,

$$\tan 23^\circ = \frac{OQ}{QP} = \frac{5.3}{QP}$$

or $QP = 12.486 \text{ cm}$

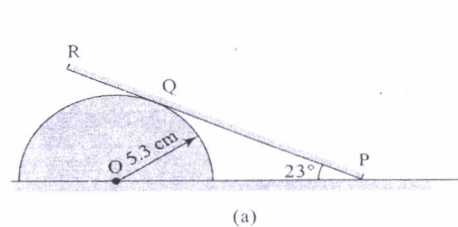


Fig. E9.46

From ΔQIP , $\angle QPI = 90^\circ - 23^\circ = 67^\circ$

Now, $\cos 67^\circ = \frac{QP}{IP} = \frac{12.486}{IP}$

So, $IP = 31.955 \text{ cm}$

Here $v_P = IP \times \omega$

or $\omega = \frac{v_P}{IP} = \frac{3.2}{31.955} = 0.1 \text{ rad/s}$

Example 9.47 A wheel is made to roll without slipping towards the right, by pulling a string wrapped around a coaxial spool as shown in Fig. E9.47(a). Determine the velocity with which the string should be pulled so that the centre of the wheel moves with a velocity of 13.5 m/s.

Solution Draw the velocity diagram with instantaneous centre of rotation. From geometry, $OA = 0.3 \text{ m}$, $OI = 0.71 \text{ m}$ [Fig. E9.47(b)]. Using similar triangles ΔIOP and ΔIAQ ,

$$\frac{v_O}{OI} = \frac{v_S}{AI} = \frac{v_S}{OI - OA}$$

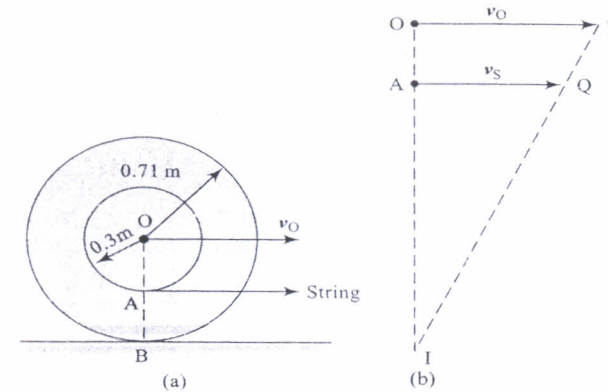
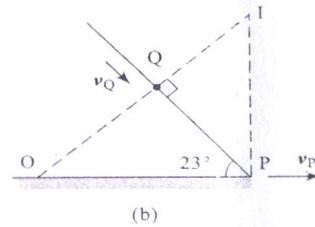


Fig. E9.47

Putting the values, we get

$$\frac{13.5}{0.71} = \frac{v_S}{0.71 - 0.3}$$

or $v_S = 7.796 \text{ m/s}$

Example 9.48 A four-bar linkage is shown in Fig. E9.48(a). Bar DC has angular velocity 4.2 rad/s counterclockwise. Determine the angular velocity of AB.

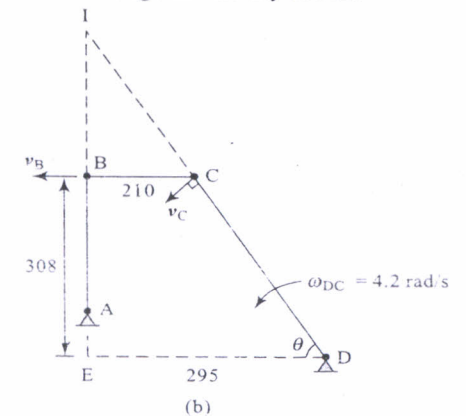
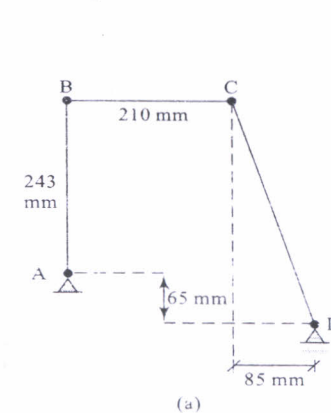


Fig. E9.48

Solution Draw the velocity diagram including instantaneous centre of rotation. Here,

$$\theta = \tan^{-1} \left(\frac{308}{85} \right) = 74.57^\circ$$

Now, $\omega_{DC} \times DC = v_C$

or $v_C = 4.2 \times \sqrt{308^2 + 85^2} = 1341.96 \text{ mm/s}$

Again from link BC, $v_C = \omega_{BC} \times BC$

or $\omega_{BC} = \frac{v_C}{BC} = \frac{1341.96}{210} = 6.39 \text{ rad/s}$

From geometry [Fig. E9.48(b)],

$$IE = DE \tan \theta = 295 \tan 74.57^\circ = 1068.8 \text{ mm}$$

$$\text{So, } IB = IE - BE = 1068.8 - 308 = 760.8 \text{ mm}$$

$$\text{Now, } v_B = \omega_{BC} \times IB = 6.39 \times 760.8 = 4861.5 \text{ mm/s}$$

From link AB,

$$v_B = \omega_{AB} \times AB$$

$$\text{or } \omega_{AB} = \frac{v_B}{AB} = \frac{4861.5}{243} = 20 \text{ rad/s}$$

Example 9.49 In the crank-shaft mechanism of a steam engine, shown in Fig. E9.49(a) has crank length 18 cm while the connecting rod BC has length 81 cm. If the rotational speed of crank AB is 320 rpm, determine the velocity and the angular speed of the connecting rod.

Solution Draw velocity triangle concerned with instantaneous centre of rotation I.

$$\text{Here } \omega_{AB} = \frac{2\pi \times 320}{60} = 33.51 \text{ rad/s}$$

$$\text{Now, } v_B = \omega_{AB} \times AB = 33.51 \times 0.18 = 6.032 \text{ m/s}$$

Applying law of sine in $\triangle ABC$ [Fig. E9.49(b)]

$$\frac{AB}{\sin \phi} = \frac{BC}{\sin 27^\circ}$$

$$\text{or } \sin \phi = \sin 27^\circ \times \frac{0.18}{0.81} = 0.1$$

$$\text{or } \phi = 5.74^\circ$$

$$\text{In } \triangle IBC, \angle IBC = 27^\circ + \phi = 32.74^\circ$$

$$\angle ICB = 90^\circ - \phi = 84.26^\circ$$

$$\angle BIC = 180^\circ - (\angle IBC + \angle ICB) = 63^\circ$$

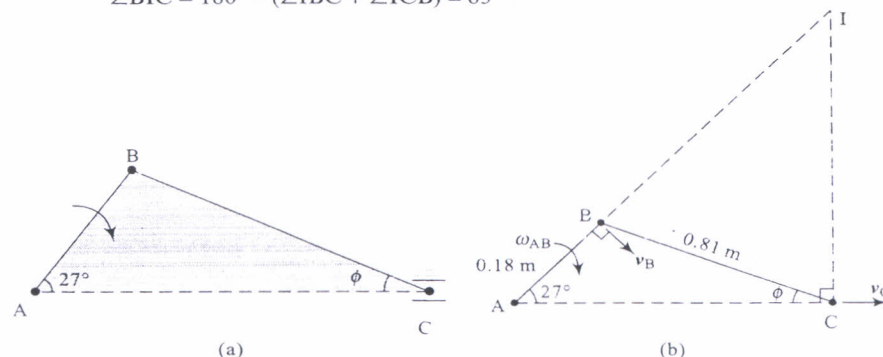


Fig. E9.49

Using law of sine,

$$\frac{BC}{\sin \angle BIC} = \frac{IB}{\sin \angle ICB}$$

$$\text{So } IB = \frac{\sin 84.26^\circ}{\sin 63^\circ} \times 0.81 = 0.9045 \text{ m}$$

$$\text{Now, } \omega_{BC} = \frac{v_B}{IB} = \frac{6.032}{0.9045} = 6.669 \text{ rad/s}$$

Example 9.50 Prove that for the arrangement shown in Fig. E9.50(a), the locus of instantaneous centre of rotation describes a circle of radius $l/(2 \sin \phi)$ touching the centre C.

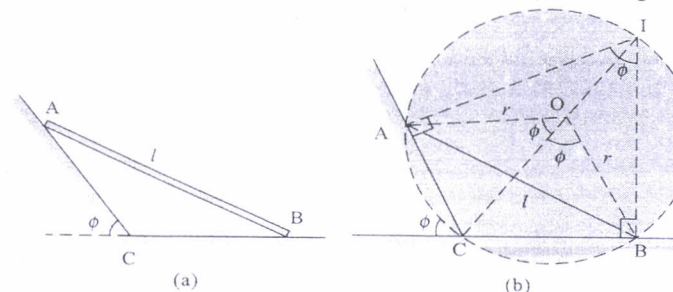


Fig. E9.50

Solution Draw the velocity diagram with instantaneous centre of rotation [Fig. E9.50 (b)]. Here ACBI is a quadrilateral having $\angle IAC = 90^\circ = \angle IBC$. If a circle is to circumscribe ACBI, touching the points I, A, C, and B, IC must be the diameter [Fig. E9.50(b)]. Also, $\angle AIB = \phi$ and $\angle AOB = 2\phi$.

Here the centre of the circumscribed circle is O and radius $OA = OC = OB = r$ (say).

From $\triangle AOB$,

$$OA \sin \phi + OB \sin \phi = AB$$

$$\text{or } r \sin \phi + r \sin \phi = l$$

$$\text{So, } r = \frac{l}{2 \sin \phi} \quad (\text{Proved})$$

Example 9.51 The velocity of centre O of a compound wheel running in no-slip condition is 2.3 m/s [Fig. E9.51(a)]. Determine the velocities at points A, D, and C.

Solution The velocity triangle identifying instantaneous centre of rotation I is drawn in Fig. E9.51(b).

Using similar triangles, $\triangle IAP$ and $\triangle IOQ$,

$$\frac{v_A}{v_O} = \frac{IA}{IO} = \frac{0.4 + 0.18}{0.18} = 3.222$$

$$\text{or } v_A = 2.3 \times 3.222 = 7.4111 \text{ m/s}$$

Again from similar triangles $\triangle IOQ$ and $\triangle ICR$,

$$\frac{v_C}{v_O} = \frac{IC}{IO} = \frac{0.4 - 0.18}{0.18} = 1.222$$

$$\text{or } v_C = 2.3 \times 1.222 = 2.8111 \text{ m/s}$$

As rotational speed of the compound wheel is constant,

$$\omega = \frac{v_O}{IO} = \left(\frac{2.3}{0.18} \right) \text{ rad/s}$$

From $\triangle IOD$, $IO = 0.18 \text{ m}$, $OD = 0.4 \text{ m}$

$$\text{So, } ID = \sqrt{0.18^2 + 0.4^2} = 0.4386 \text{ m}$$

$$\text{Hence, } v_D = \omega \times ID = \frac{2.3}{0.18} \times 0.4386 = 5.604 \text{ m/s}$$

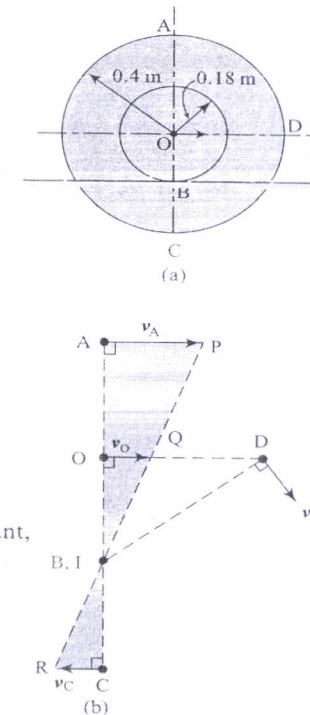


Fig. E9.51

Example 9.52 For the instant represented, as shown in Fig. E9.52(a), when the crank PQ passes the horizontal position, determine the velocity of the centre C of the link QR by the method of instantaneous centre.

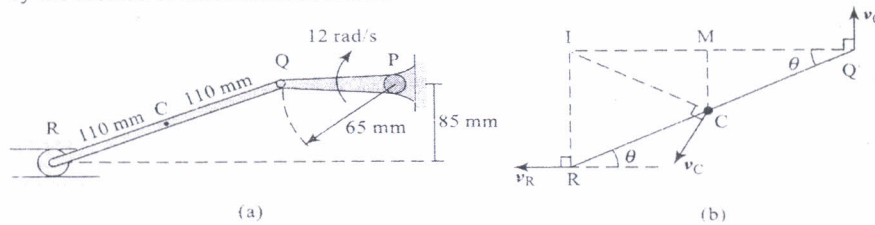


Fig. E9.52

Solution Draw the velocity diagram considering instantaneous centre of rotation I [Fig. E9.52(b)]

From geometry, $\theta = \sin^{-1} \left(\frac{0.085}{0.220} \right) = 22.728^\circ$

Now, $v_Q = PQ \times \omega_{PQ} = 0.065 \times 12 = 0.78 \text{ m/s}$

Again, $v_Q = \omega_{RQ} \times IQ$

or $\omega_{RQ} = \frac{v_Q}{RQ \cos \theta} = \frac{0.78}{0.22 \cos 22.728^\circ} = 3.844 \text{ rad/s}$

From $\triangle CIM$,

$$IM = \frac{1}{2} IQ = \frac{1}{2} RQ \cos 22.728^\circ = 0.1015 \text{ m}$$

and $CM = \frac{1}{2} IR = \frac{0.085}{2} = 0.0425 \text{ m}$

So, $IC = (IM^2 + CM^2)^{1/2} = 0.11 \text{ m}$

Here, $\omega_C = \omega_{RQ} = 3.844 \text{ rad/s}$

Hence, $v_C = \omega_C \times IC = 3.844 \times 0.11 = 0.4228 \text{ m/s}$

Example 9.53 A linkage system is shown in Fig. E9.53(a). At the instant represented here, determine the velocity of the guide roller R and of C, midway of Q and R.

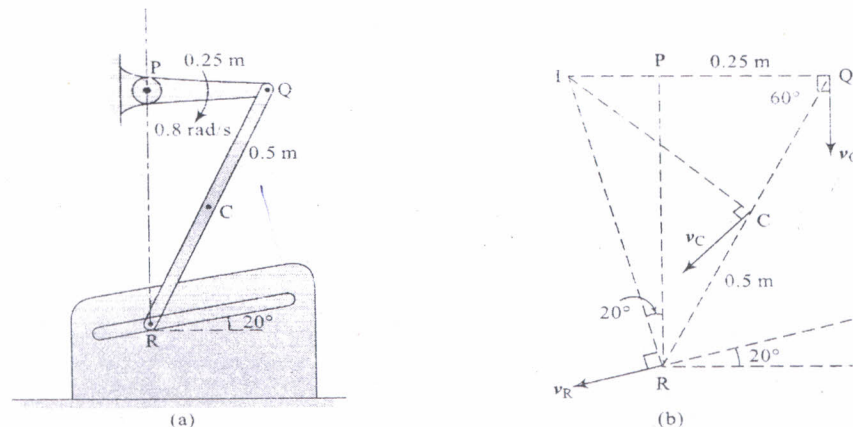


Fig. E9.53

Solution Draw the velocity diagram and find the location of instantaneous centre of rotation [Fig. E9.53(b)].

Now, $PR = \sqrt{0.5^2 - 0.25^2} = 0.433 \text{ m}$

Again $IR \cos 20^\circ = PR$ or $IR = 0.4608$

Here, $v_Q = \omega_{PQ} \times PQ = 0.8 \times 0.25 = 0.2 \text{ m/s}$

Considering the bar RQ, $v_Q = \omega_{RQ} \times RQ$

So, $\omega_{RQ} = \frac{v_Q}{RQ} = \frac{0.2}{0.5} = 0.4 \text{ rad/s}$

Considering the roller R,

$$v_R = \omega_{RQ} \times IR = 0.4 \times 0.4608 = 0.18432 \text{ m/s}$$

Now, $\angle PQC = \cos^{-1} \left(\frac{0.25}{0.5} \right) = 60^\circ$

and $IP = IR \sin 20^\circ = 0.1576 \text{ m}$

In $\triangle IQC$, $IQ = 0.1576 + 0.25 = 0.4076 \text{ m}$, $QC = 0.25 \text{ m}$, and $\angle IQC = 60^\circ$

So, $IC = [IQ^2 + QC^2 - 2IQ \times QC \times \cos \angle IQC]^{1/2}$
 $= (0.4076^2 + 0.25^2 - 2 \times 0.4076 \times 0.25 \times \cos 60^\circ)^{1/2} = 0.356 \text{ m}$

Hence, $v_C = \omega_{RQ} \times IC = 0.4 \times 0.356 = 0.1424 \text{ m/s}$

Example 9.54 A mechanism is shown in Fig. E9.54(a). For an interval of its motion, the piston rod of hydraulic cylinder has a velocity $v_P = 1.5 \text{ m/s}$. At a certain instant, $\theta = \beta = 60^\circ$. By the method of instantaneous centre, determine the angular velocity ω_{QR} of link QR.

Solution Draw the velocity diagram considering centre of instantaneous rotation [Fig. E9.54(b)].

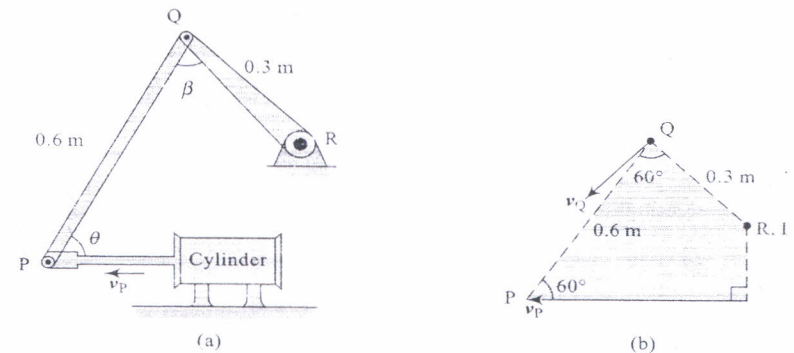


Fig. E9.54

Here, rotational speed of link PQ,

$$\omega_{PQ} = \frac{v_P}{PQ} = \frac{1.5}{0.6} = 2.5 \text{ rad/s}$$

Again considering the motion of Q,

$$v_Q = \omega_{QR} \times QR = \omega_{PQ} \times PQ$$

or $\omega_{QR} = \omega_{PQ} \times \frac{PQ}{QR} = 2 \times 2.5 = 5.0 \text{ rad/s}$